

The diencephalon is part of the prosencephalon (forebrain), which develops from the foremost primary cerebral vesicle and differentiates into a caudal diencephalon and rostral telencephalon. The cerebral hemisphere develops from the side of the telencephalon, containing a lateral ventricle. The sites of evagination become the interventricular foramina, through which the two lateral ventricles and midline third ventricle communicate. The diencephalon corresponds largely to the structures that develop lateral to the third ventricle. The lateral walls of the diencephalon form the epithalamus most superiorly, the thalamus centrally, and the subthalamus and hypothalamus inferiorly.

THALAMUS

The thalamus plays a crucial role in many brain functions, serving as a processing and distribution centre, relaying and regulating information from the outside world and the internal milieu to the cerebral cortex and vice versa. It is involved in multiple activities, including consciousness, sleep, memory and sensory and motor functions. Our increasing understanding of the role of the thalamus is providing insights into pathological disorders of the brain and is opening up the possibility of targeting its various constituent nuclei to treat a variety of disorders, including epilepsy, Parkinson's disease, pain and psychiatric disorders (Table 23.1; see Fig. 23.5) (For review, see Lozano and Lipsman 2013.)

Morphologically, the thalamus is a large ovoid nuclear mass, about 4 cm long, which borders the dorsal part of the third ventricle (Figs 23.1–23.2). Its narrow anterior pole lies close to the midline and forms the posterior boundary of the interventricular foramen. Posteriorly, an expansion, the pulvinar, extends beyond the third ventricle to overhang the superior colliculus (see Fig. 21.5). The brachium of the superior colliculus (superior quadrigeminal brachium) separates the pulvinar above from the medial geniculate body below. A small oval elevation, the lateral geniculate body, lies lateral to the medial geniculate.

The superior (dorsal) surface of the thalamus is covered by a thin layer of white matter, the stratum zonale. This curved surface is separated from the overlying body of the fornix by the choroid fissure with the tela choroidea within it. More laterally, it forms part of the floor of the lateral ventricle. The lateral border of the superior surface of the thalamus is marked by the stria terminalis and overlying thalamostriate vein, which separate the thalamus from the body of the caudate nucleus. Laterally, a slender sheet of white matter, the external medullary lamina, separates the main body of the thalamus from the reticular nucleus. Lateral to this, the thick posterior limb of the internal capsule lies between the thalamus and the lentiform complex.

The medial surface of the thalamus is the superior (dorsal) part of the lateral wall of the third ventricle. It is usually connected to the

contralateral thalamus by an interthalamic adhesion (Fig. 18.10). The boundary with the hypothalamus is marked by a faint hypothalamic sulcus, which curves from the upper end of the cerebral aqueduct to the interventricular foramen. The thalamus is continuous with the mid-brain tegmentum, the subthalamus and the hypothalamus.

Internally, the greater part of the thalamus is divided into anterior, medial and lateral nuclear groups by a vertical Y-shaped sheet of white matter, the internal medullary lamina (Figs 23.3–23.4; Table 23.2). In addition, intralaminar nuclei lie embedded within the internal medullary lamina and midline nuclei about the lateral walls of the third ventricle. The reticular nucleus forms a shell-like lateral covering to the main nuclear mass, separated from it by an external medullary lamina of nerve fibres.

In general, thalamic nuclei both project to and receive fibres from the cerebral cortex (Jones 1985, Macchi and Jones 1997) (see Fig. 23.4). The whole cerebral cortex, including neocortex, paleocortex of the piriform lobe and archicortex of the hippocampal formation, is reciprocally connected with the thalamus. The thalamus is the major route by which subcortical neuronal activity influences the cerebral cortex, and the greatest input to most thalamic nuclei comes from the cerebral cortex.

The projection to the thalamus from the cortex is precisely reciprocal; each cortical area projects in a topographically organized manner to all sites in the thalamus from which it receives an input (see Table 23.2). Thalamocortical fibres terminate predominantly in layer IV of the cortex. Corticothalamic fibres that reciprocate 'specific' thalamocortical pathways arise from modified pyramidal cells of layer VI, whereas those reciprocating 'non-specific' inputs arise from typical pyramidal cells of layer V, and may in part be axon collaterals of other cortico-subcortical pathways.

It is customary to consider thalamic nuclei as either 'specific' nuclei, which mediate finely organized and precisely transmitted sensory information to discrete cortical sensory areas, or as 'non-specific' nuclei, with less precisely defined functions.

The specific nuclei are further subdivided into relay nuclei and association nuclei. However, many nuclei classified as specific may also send non-specific projections to widespread cortical areas. Similarly, the division of thalamic nuclei into relay and association groups rests on the assumption that relay nuclei receive a major subcortical pathway, whereas association nuclei are reciprocally connected to cortical areas that are traditionally referred to as association areas and which receive their principal non-cortical input from other subcortical structures.

Table 23.1 Thalamic deep brain stimulation targets

Indication	Target	References
Pain	VPI/VPm, PVG	Boccard et al 2013
Tremor	VL (Vim)	Benabid et al 1991, Hubble et al 1996
Epilepsy	Anterior thalamus	Hodaie et al 2002
Obsessive–compulsive disorder	Inferior thalamic peduncle	Jiménez et al 2013
Minimally conscious state	CM (central thalamus)	Giacino et al 2012
Tourette's syndrome	CM	Piedad et al 2012
Dystonia	VL (Voa/Vop), VL (Vim)	Cooper 1976, Tasker et al 1988
Myoclonus–dystonia	Vim	Gruber et al 2010

Abbreviations: CM, centromedian nucleus of thalamus; PVG, periventricular grey; Vim, ventral intermediate nucleus of thalamus (nucleus ventrointermedius); VL, ventral lateral thalamic nucleus (nucleus ventrolateralis); Voa, nucleus ventro-oralis anterior; Vop, nucleus ventro-oralis posterior; VPI, ventral posterolateral nucleus; VPm, ventral posteromedial nucleus.

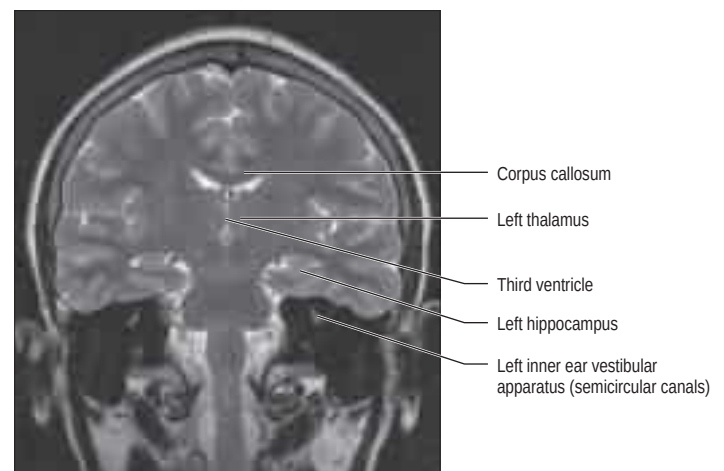


Fig. 23.1 A coronal T2-weighted magnetic resonance image at the level of the thalamus and third ventricle. (Courtesy of Professor Alan Jackson, Wolfson Molecular Imaging Centre, The University of Manchester, Manchester, UK.)

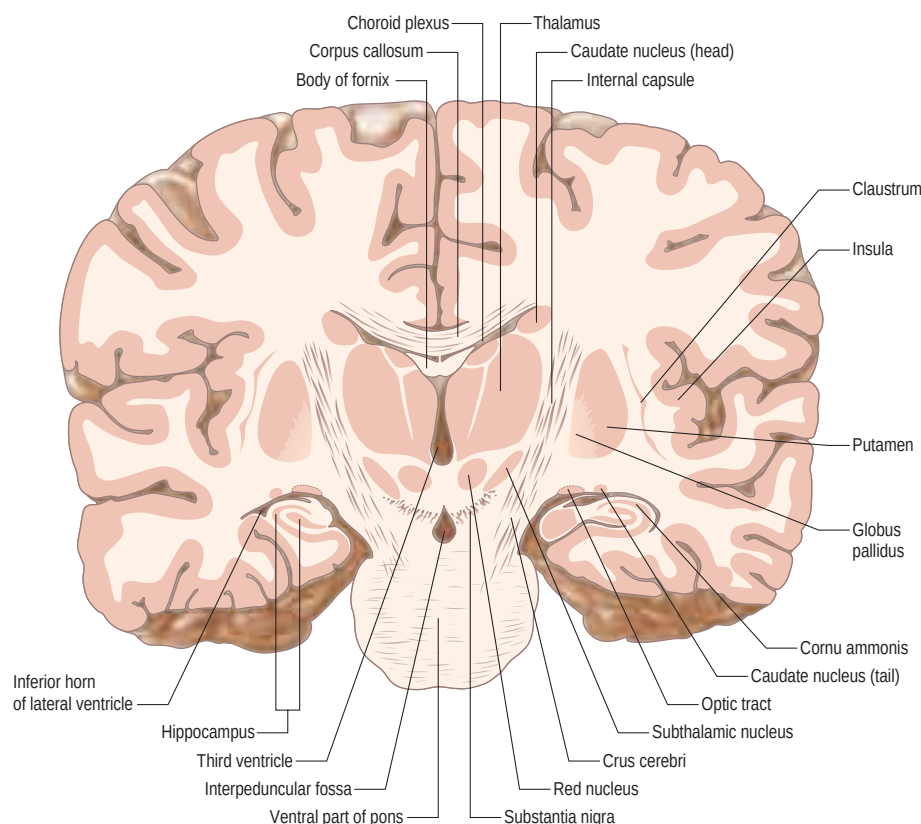


Fig. 23.2 The principal parts of the diencephalon and basal ganglia, coronal section.

Table 23.2 Main thalamic nuclei and their major afferent and efferent connections

Group	Nucleus	Major subcortical connection	Major cortical connections	System
Anterior	Anterior	Mammillary bodies (MTT)	Cingulate and parahippocampal gyrus	Relay limbic
Medial	Dorsomedial	Amygdala, thalamic nuclei	Prefrontal cortex	Association
Lateral				
Ventral	VA	GPI, SNr	Premotor cortex	Relay motor
	VL	Ipsilateral GPI, contralateral cerebellum	Supplementary and primary motor cortex	Relay motor
	VPI, VPm	STT to VPI, TTT to VPm, MLT to anterior surface	Somatosensory cortex, insula	Relay sensory
	LGB	Optic tract	Visual cortex	Visual
	MGB	Auditory pathway (inferior brachium)	Auditory cortex	Auditory
Dorsal	Pulvinar – LP	Superior colliculus	Parieto-occipito-temporal	Association
	LD	Pretectum and superior colliculus	Cingulate, parahippocampus	
Intralaminar	CM	GPI, cerebellum	Striatum, motor cortex	
Reticular		Widespread connections	Widespread connections	

Abbreviations: CM, centromedian nucleus of thalamus; GPI, internal segment of globus pallidus; LD, lateral dorsal nucleus of thalamus; LGB, lateral geniculate body; LP, lateral posterior nucleus of thalamus; MGB, medial geniculate body; MLT, medial lemniscal tract; MTT, mammillothalamic tract; SNr, substantia nigra pars reticularis; STT, spinothalamic tract; TTT, trigeminothalamic tract; VA, ventral anterior nucleus; VL, ventral lateral nucleus; VPI, ventral posterolateral nucleus; VPm, ventral posteromedial nucleus.

There is little evidence of significant intrathalamic connectivity, but there are increasing indications of non-cortical afferent pathways linked to so-called association nuclei; the extensive connectivity between the reticular nucleus and other thalamic nuclei is a notable exception.

ANTERIOR THALAMIC NUCLEI

The anterior group of nuclei lie between the arms of the Y-shaped internal medullary lamina, and underlie the anterior thalamic tubercle (see Figs 21.5, 23.4). Three subdivisions are recognized: the largest is the anteroventral nucleus, the others being the anteromedial and anterodorsal nuclei.

The anterior nuclei are the principal recipients of the mammillothalamic tract, which arises from the mammillary nuclei of the hypothalamus.

The mammillary nuclei receive fibres from the hippocampal formation via the fornix. The medial mammillary nucleus projects to the ipsilateral anteroventral and anteromedial thalamic nuclei, and the lateral mammillary nucleus projects bilaterally to the anterodorsal nuclei. The nuclei of the anterior group also receive a prominent cholinergic input from the basal forebrain and the brainstem.

The cortical targets of efferent fibres from the anterior nuclei of the thalamus lie largely on the medial surface of the hemisphere. They include the anterior limbic area (in front of and inferior to the corpus callosum), the cingulate gyrus and the parahippocampal gyrus (including the medial entorhinal cortex and the presubiculum and parasubiculum). There also appear to be minor connections between the anterior nuclei and the dorsolateral prefrontal and posterior areas of neocortex. The anterior thalamic nuclei are believed to be involved in the regulation of alertness and attention and in the acquisition of memory.

MEDIAL THALAMIC NUCLEI

The single component of this thalamic region is the mediodorsal or dorsomedial nucleus, which is particularly large in humans. Laterally, it is limited by the internal medullary lamina and intralaminar nuclei. Medially, it abuts the midline parataenial and reuniens (medioventral) nuclei. It is divided into anteromedial magnocellular and posterolateral parvocellular parts.

The smaller magnocellular division receives olfactory input from the piriform and adjacent cortex, the ventral pallidum, and the amygdala.

The mediobasal amygdaloid nucleus projects to the dorsal part of the anteromedial magnocellular nucleus, and the lateral nuclei project to the more central and anteroventral regions. The anteromedial magnocellular nucleus projects to the anterior and medial prefrontal cortex, notably to the lateral posterior and central posterior olfactory areas on the orbital surface of the frontal lobe. In addition, fibres pass to the ventromedial cingulate cortex and some to the inferior parietal cortex and anterior insula. These cortical connections are reciprocal.

The larger, posterolateral parvocellular division connects reciprocally with the dorsolateral and dorsomedial prefrontal cortex, the anterior cingulate gyrus and the supplementary motor area. In addition, efferent fibres pass to the posterior parietal cortex.

The mediodorsal nucleus appears to be involved in a wide variety of higher functions. Damage may lead to a decrease in anxiety, tension, aggression or obsessive thinking. There may also be transient amnesia,

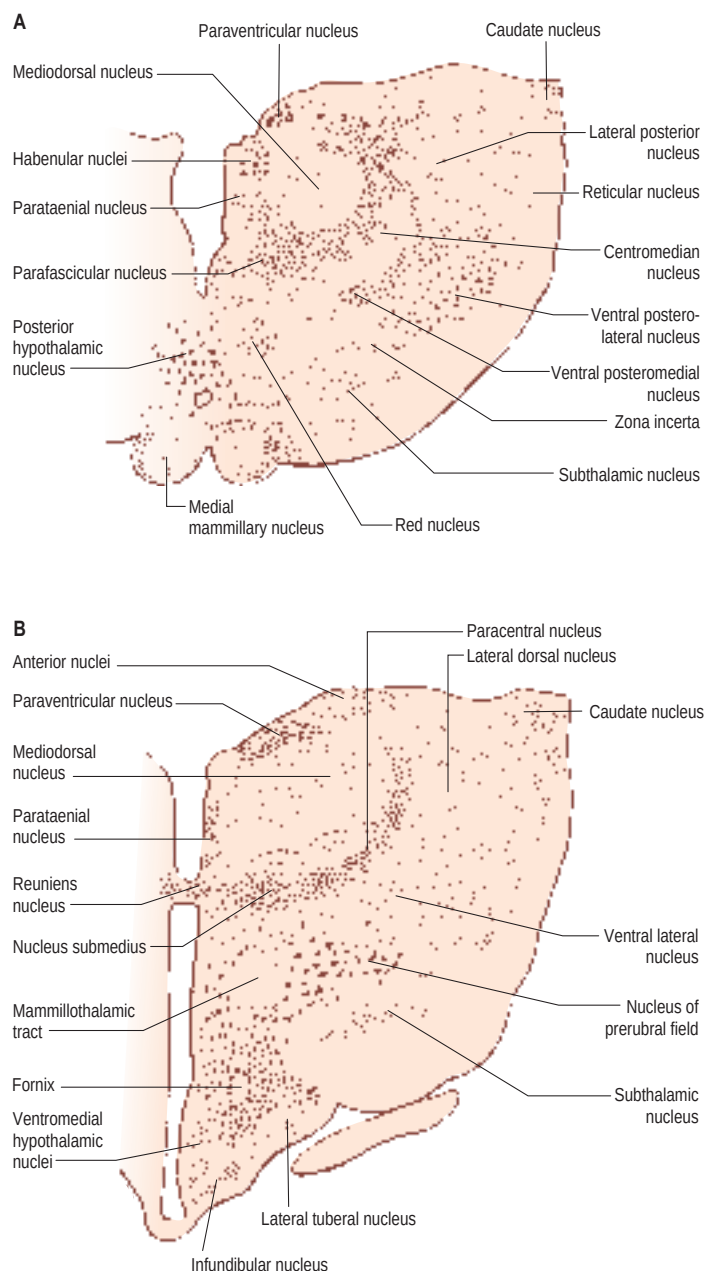


Fig. 23.3 The main nuclear aggregations of cell bodies in the diencephalon, coronal sections. **A**, At the level of the mammillary bodies. **B**, At the level of the tuber cinereum. Note the variations in cell size, shape and packing density, which characterize the nuclear masses of the thalamus, subthalamus and hypothalamus at these levels.

with confusion developing particularly over the passage of time. Much of the neuropsychology of medial nuclear damage reflects defects in functions similar to those performed by the prefrontal cortex, with which it is closely linked. The effects of ablation of the mediodorsal nuclei parallel, in part, the results of prefrontal lobotomy.

LATERAL THALAMIC NUCLEI

The lateral nuclear complex, lying lateral to the internal medullary lamina, is the largest division of the thalamus. It is divided into dorsal and ventral tiers of nuclei. The lateral dorsal nucleus, lateral posterior nucleus and the pulvinar all lie dorsally. The lateral and medial geniculate nuclei lie inferior to the pulvinar near the posterior pole of the thalamus. The ventral-tier nuclei are the ventral anterior, ventral lateral and ventral posterior nuclei.

Ventral anterior nucleus

The ventral anterior (VA) nuclear complex lies at the anterior pole of the ventral nuclear group. It is limited anteriorly by the reticular nucleus and posteriorly by the ventral lateral nucleus, and lies between the external and internal medullary laminae. It consists of a principal part

(VAp) and a magnocellular part (VAMc). The subcortical connections to this region are largely ipsilateral from the internal pallidum and the pars reticularis of the substantia nigra. The terminal fields from these origins do not overlap. Fibres from the globus pallidus end in the principal part of the ventral anterior nuclear complex. The substantia nigra projects to the magnocellular part of the ventral anterior nuclear complex. Corticothalamic fibres from the premotor cortex (area 6) terminate in the principal part and fibres from the frontal eye field (area 8) terminate in the magnocellular part. The ventral anterior thalamus does not appear to receive fibres directly from the motor cortex. The efferent projections from the ventral anterior nuclear complex are incompletely known. Some pass to intralaminar thalamic nuclei and others project to widespread regions of the frontal lobe and to the anterior parietal cortex. Their functions are unclear. The ventral anterior thalamus appears to play a central role in the transmission of the cortical 'recruiting response', a phenomenon in which stimulation of the thalamus can initiate long-lasting, high-voltage, repetitive negative electrical waves over much of the cerebral cortex.

Ventral lateral nucleus

The ventral lateral (VL) nucleus consists of two major divisions with distinctly different connections and functions. The anterior division, or pars oralis (VLO), receives topographically organized fibres from the ipsilateral internal pallidum. The posterior division, or pars caudalis (VLC), receives topographically organized fibres from the contralateral deep cerebellar nuclei. Additional subcortical projections have been reported from the spinothalamic tract and the vestibular nuclei. Numerous cortical afferents to both the pars oralis and the pars caudalis originate from precentral motor cortical areas, including both area 4 and area 6.

The pars oralis of the ventral lateral nucleus sends efferent fibres to the supplementary motor cortex on the medial surface of the hemisphere and to the lateral premotor cortex. The pars caudalis of the ventral lateral nucleus projects efferent fibres to the primary motor cortex, where they end in a topographically arranged fashion. The head region of area 4 receives fibres from the medial part of pars caudalis, and the leg region receives fibres from the lateral pars caudalis.

Approximately 70% of ventral lateral neurones are large relay neurones, and the other 30% are local circuit interneurones. Responses can be recorded in ventral lateral thalamic neurones during both passive and active movement of the contralateral body. The topography of its connections, and recordings made within the nucleus, suggest that the pars caudalis contains a body representation comparable to that in the ventral posterior nucleus. In patients with tremor, clusters of ventral lateral neurones fire in bursts that are synchronous with peripheral tremor: these cells have been designated as 'tremor cells'.

Stereotaxic surgery of the ventral lateral nucleus is sometimes used in the treatment of essential tremor (see Table 23.1; Fig. 23.5). In the past, thalamotomy was used extensively for the treatment of Parkinson's disease. The internal pallidum and the subthalamic nucleus are now the preferred neurosurgical targets for Parkinson's disease.

Ventral posterior nucleus

The ventral posterior (VP) nucleus is the principal thalamic relay for the somatosensory pathways. It is thought to consist of two major divisions: the ventral posterolateral (VPL) and ventral posteromedial (VPM) nuclei. The ventral posterolateral nucleus receives the medial lemniscal and spinothalamic pathways, and the ventral posteromedial nucleus receives the trigeminothalamic pathway. Connections from the vestibular nuclei and lemniscal fibres terminate along the ventral surface of the ventral posterior nucleus.

There is a well-ordered topographic representation of the body in the ventral posterior nucleus. The ventral posterolateral nucleus is organized so that sacral segments are represented laterally and cervical segments medially. The latter about the face area of representation (trigeminal territory) in the ventral posteromedial nucleus. Taste fibres synapse most anteriorly and ventromedially within the ventral posterolateral nucleus.

At a more detailed level, single body regions are represented as curved lamellae of neurones, parallel to the lateral border of the VP nucleus, such that there is a continuous overlapping progression of adjacent receptive fields from dorsolateral to ventromedial. Considerably less change in location of receptive field on the body is seen when passing anteroposteriorly through the nucleus. While not precisely dermatomal in nature, these curvilinear lamellae of cells probably derive from afferents related to a few adjacent spinal segments. There is

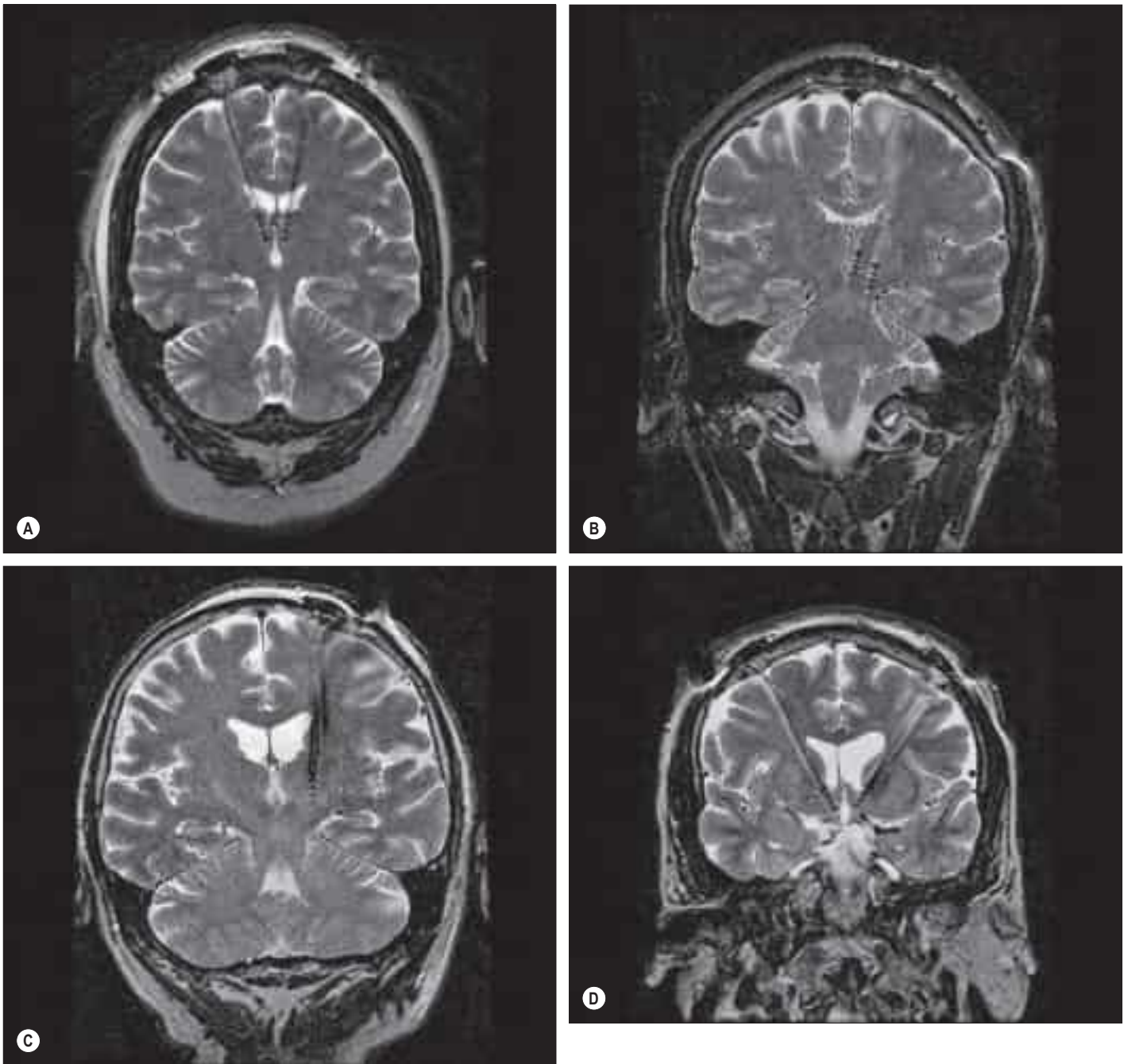


Fig. 23.5 Magnetic resonance images from four different patients with neurological or psychiatric disorders, showing the position of deep brain-stimulating (DBS) electrodes. **A**, Bilateral anterior thalamic DBS for epilepsy. **B**, Left thalamic DBS for pain, with electrodes in the sensory relay nucleus (ventral caudal nucleus of the thalamus) laterally and the periventricular grey medially. **C**, DBS electrode in the ventral intermediate nucleus of the thalamus in a patient with essential tremor. **D**, Bilateral inferior thalamic peduncle DBS electrodes in a patient with obsessive-compulsive disorder.

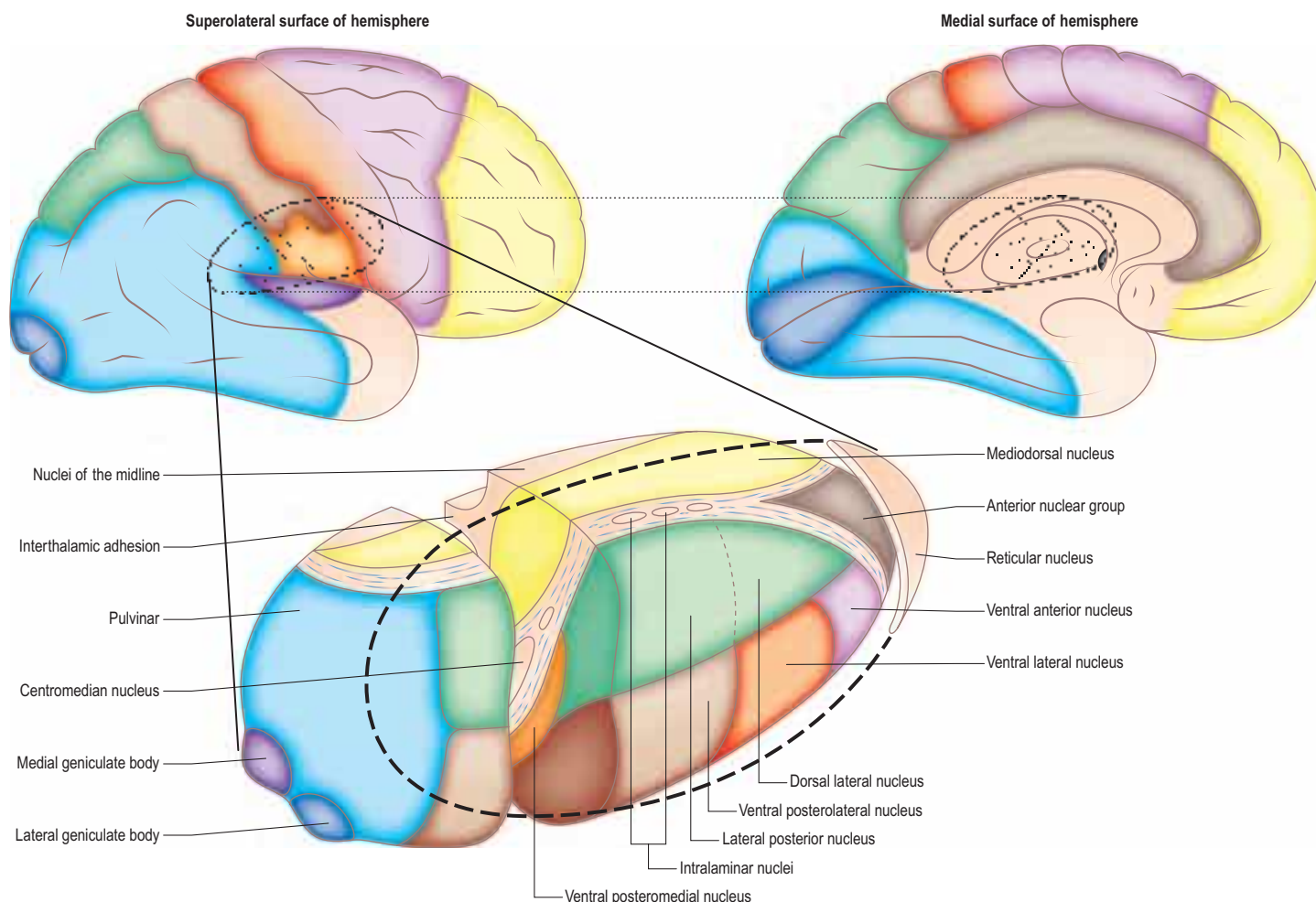


Fig. 23.4 The main nuclear masses of the thalamus (viewed from the lateral aspect in the lower illustration), colour-coded to indicate the areas of cerebral neocortex with which they are interconnected. The lack of colour in the centromedian, intralaminar and reticular nuclei and in areas of the frontal and temporal lobes is not related to the colour code. The reticular nucleus lies lateral to the main mass of the thalamus. Only the anterior pole of the reticular nucleus is shown, its posterior extent being depicted by the heavy interrupted line.

considerable distortion of the body map within the nucleus, reflecting the differences in the density of peripheral innervation that occur in different body regions, e.g. many more neurones respond to stimulation of the hand than of the trunk. Within a single lamella, neurones in the anterodorsal part of the nucleus respond to deep stimuli, including movement of joints, tendon stretch, and manipulation of muscles. Most ventrally, neurones once again respond to deep stimuli, particularly tapping. Intervening cells within a single lamella respond only to cutaneous stimuli. This organization has been confirmed by recordings made in the human ventral posterior nucleus.

The ventral posterior nucleus projects to the primary somatic sensory cortex (SI) of the postcentral gyrus and to the second somatic sensory area (SII) in the parietal operculum. The posterior part of the ventral posteromedial nucleus projects to the insular cortex. Within the primary sensory cortex, the central cutaneous core of the ventral posterior nucleus projects solely to area 3b; dorsal and ventral to this, a narrow band of cells projects to both area 3b and area 1. The most dorsal and ventral deep stimulus receptive cells project to areas 3a and 2. The whole nucleus projects to the second somatic sensory area.

Medial geniculate nucleus

The medial geniculate nucleus, which is a part of the auditory pathway, is located within the medial geniculate body, a rounded elevation situated posteriorly on the ventrolateral surface of the thalamus, and separated from the pulvinar by the superior quadrigeminal brachium. It receives fibres travelling in the inferior quadrigeminal brachium. Three major subnuclei – medial, ventral and dorsal – are recognized within it. The inferior brachium separates the medial (magnocellular) nucleus, which consists of sparse, deeply staining neurones, from the lateral nucleus, which is made up of medium-sized, densely packed and darkly staining cells. The dorsal nucleus overlies the ventral nucleus and expands posteriorly; it is, therefore, sometimes known as the posterior nucleus of the medial geniculate. It contains small- to medium-sized,

pale-staining cells, which are less densely packed than those of the lateral nucleus. The ventral nucleus receives fibres from the central nucleus of the ipsilateral inferior colliculus via the inferior quadrigeminal brachium and also from the contralateral inferior colliculus. The nucleus contains a complete tonotopic representation. Low-pitched sounds are represented laterally, and progressively higher-pitched sounds are encountered as the nucleus is traversed from lateral to medial. The dorsal nucleus receives afferents from the pericentral nucleus of the inferior colliculus and from other brainstem nuclei of the auditory pathway. A tonotopic representation has not been described in this subdivision and cells within the dorsal nucleus respond to a broad range of frequencies. The magnocellular medial nucleus receives fibres from the inferior colliculus and from the deep layers of the superior colliculus. Neurones within the magnocellular subdivision may respond to modalities other than sound. However, many cells respond to auditory stimuli, usually to a wider range of frequencies than neurones in the ventral nucleus. Many units show evidence of binaural interaction, with the leading effect arising from stimuli in the contralateral cochlea. The ventral nucleus projects primarily to the primary auditory cortex. The dorsal nucleus projects to auditory areas surrounding the primary auditory cortex. The magnocellular division projects diffusely to auditory areas of the cortex and to adjacent insular and opercular fields.

Lateral geniculate nucleus

The lateral geniculate body, which is part of the visual pathway, is a small, ovoid, ventral projection from the posterior thalamus (see Figs 21.5, 23.4). The superior quadrigeminal brachium enters the posteromedial part of the lateral geniculate body dorsally, lying between the medial geniculate body and the pulvinar.

The lateral geniculate nucleus is an inverted, somewhat flattened, U-shaped nucleus and is laminated (Fig. 23.6A). Its internal organization is usually described on the basis of six laminae, although seven or



Single lemniscal axons have an extended anteroposterior terminal zone within the nucleus. Rods of cells running the length of the anteroposterior, dorsoventrally orientated, lamellae respond with closely similar receptive field properties and locations, derived from a small bundle of lemniscal afferents. It appears, therefore, that each lamella contains the complete representation of a single body part, e.g. a finger. Lamellae consist of multiple narrow rods of neurones, orientated anteroposteriorly, each of which receives input from the same small region of the body that is represented within the lamella, and from the same type of receptors. These thalamic 'rods' form the basis for both place- and modality-specific input to columns of cells in the somatic sensory cortex. Spinothalamic tract afferents to the ventral posterolateral nucleus terminate throughout the nucleus. The neurones from which these axons originate appear to be mainly of the 'wide-dynamic-range' class, responding to both low-threshold mechanoreceptors and high-threshold nociceptors; a smaller proportion of neurones respond exclusively to high-threshold nociceptors. Some neurones respond to temperature changes. There is evidence that spinothalamic tract neurones carrying nociceptive and thermal information terminate in a distinct nuclear area, identified as the posterior part of the ventral medial nucleus (VMpo).

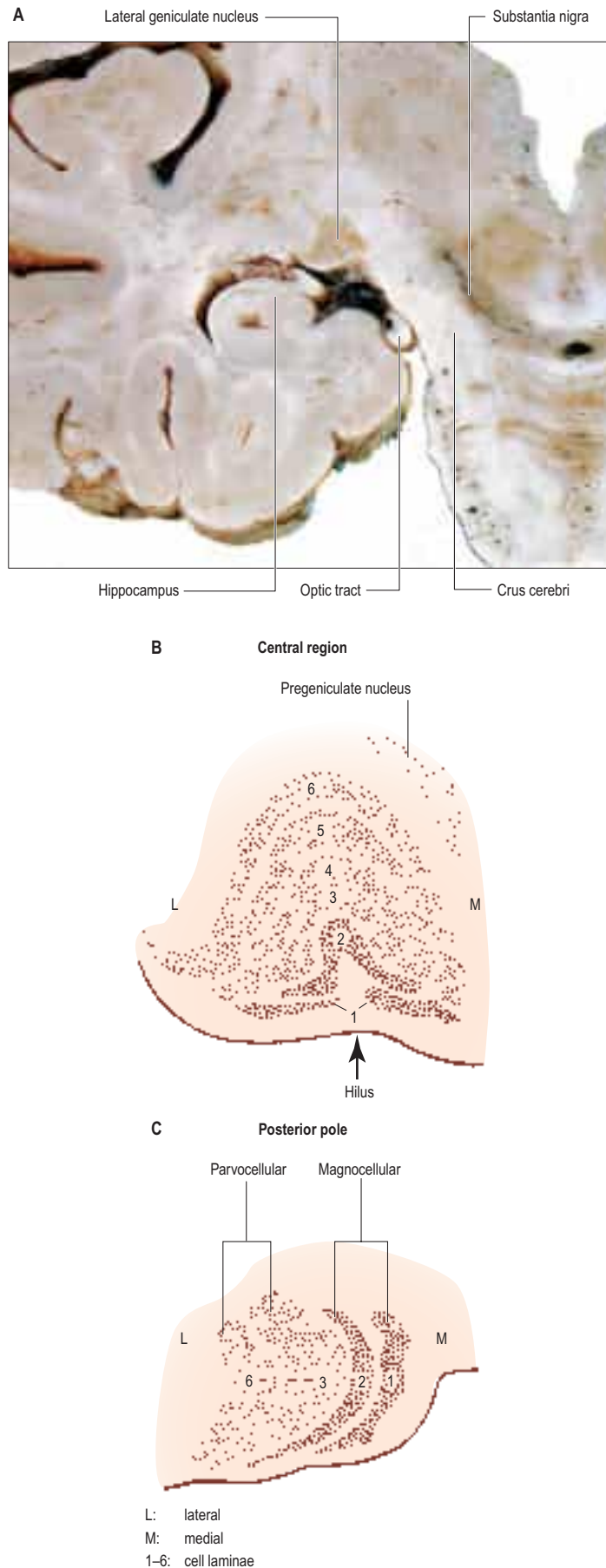


Fig. 23.6 **A**, A coronal section showing the lateral geniculate nucleus. **B–C**, Coronal sections through the lateral geniculate nucleus illustrating the laminar arrangement of neurones (**B**) near its central region and (**C**) near its posterior pole. (**A**, Photograph by Kevin Fitzpatrick, London.)

eight may be present. The laminae are numbered 1–6, from the innermost ventral to the outermost dorsal (**Fig. 23.6BC**). Laminae 1 and 2 consist of large cells, the magnocellular layers, whereas layers 4–6 have smaller neurones, the parvocellular laminae. The apparent gaps between laminae are called the interlaminar zones. Most ventrally, an additional superficial, or S, lamina is recognized.

The lateral geniculate nucleus receives a major afferent input from the retina. The contralateral nasal hemiretina projects to laminae 1, 4 and 6, whereas the ipsilateral temporal hemiretina projects to laminae 2, 3 and 5. The parvocellular laminae receive axons predominantly of X-type retinal ganglion cells, i.e. more slowly conducting cells with sustained responses to visual stimuli. The faster-conducting, rapidly adapting Y-type retinal ganglion cells project mainly to the magnocellular laminae 1 and 2, and give off axonal branches to the superior colliculus. A third type of retinal ganglion cell, the W cells, which have large receptive fields and slow responses, project to both the superior colliculus and the lateral geniculate nucleus, where they terminate particularly in the interlaminar zones and in the S lamina.

The lateral geniculate nucleus is organized in a visuotopic manner and it contains a precise map of the contralateral visual field. The vertical meridian is represented posteriorly, the peripheral anteriorly, the upper field laterally, and the lower field medially. Similar precise, point-to-point representation is also found in the projection of the lateral geniculate nucleus to the visual cortex. Radially arranged inverted pyramids of neurones in all laminae respond to a single small area of the contralateral visual field and project to a circumscribed area of cortex. The termination of geniculocortical axons in the visual cortex is considered in detail elsewhere.

The efferent fibres of the lateral geniculate nucleus pass principally to the primary visual cortex (area 17) in the banks of the calcarine sulcus. It is possible that additional small projections pass to extrastriate visual areas in the occipital lobe, possibly arising primarily in the interlaminar zones. For further reading on processing in the lateral geniculate nucleus, see [Casagrande and Ichida \(2011\)](#).

Lateral dorsal nucleus

The lateral dorsal nucleus is the most anterior of the dorsal tier of lateral nuclei. Its anterior pole lies within a splitting of the internal medullary lamina, and posteriorly it merges with the lateral posterior nucleus. Subcortical afferents to the lateral dorsal nucleus are from the pretectum and superior colliculus. It is connected with the cingulate, retrosplenial and posterior parahippocampal cortices, the presubiculum of the hippocampal formation, and the parietal cortex.

Lateral posterior nucleus

The lateral posterior nucleus, which lies dorsal to the ventral posterior nucleus, receives its subcortical afferents from the superior colliculus. It is reciprocally connected with the superior parietal lobe. Additional connections have been reported with the inferior parietal, cingulate and medial parahippocampal cortex.

Pulvinar

The pulvinar corresponds to the posterior expansion of the thalamus, which overhangs the superior colliculus (see **Fig. 21.5**). It has three major subdivisions, which are the medial, lateral and inferior pulvinar nuclei. The medial pulvinar nucleus is dorsomedial and consists of compact, evenly spaced neurones. The inferior pulvinar nucleus lies laterally and inferiorly, and is traversed by bundles of axons in the mediolateral plane, an arrangement that confers a fragmented appearance of horizontal cords or sheets of cells separated by fibre bundles. The inferior pulvinar nucleus lies most inferiorly and laterally, and is a more homogeneous collection of cells.

The subcortical afferents to the pulvinar are uncertain. Medial and lateral pulvinar nuclei may receive fibres from the superior colliculus. It has been suggested that the inferior pulvinar nucleus receives fibres both from the superior colliculus and directly from the retina, and that it contains a complete retinotopic representation.

The cortical targets of efferent fibres from the pulvinar are widespread. In essence, the medial pulvinar nucleus projects to association areas of the parietotemporal cortex, whereas lateral and inferior pulvinar nuclei project to visual areas in the occipital and posterior temporal lobes. Thus, the inferior pulvinar nucleus connects with the striate and extrastriate cortex in the occipital lobe, and with visual association areas in the posterior part of the temporal lobe. The lateral pulvinar nucleus connects with extrastriate areas of the occipital cortex, with

Aside from retinal afferents, the lateral geniculate nucleus receives a major corticothalamic projection, the axons of which ramify densely in the interlaminar zones. The major part of this projection arises from the primary visual cortex, Brodmann's area 17, but smaller projections from extrastriate visual areas pass to the magnocellular and S laminae. Other afferents include: fibres from the superficial layer of the superior colliculus (which terminate in the interlaminar zone between laminae 1 and 2, and 2 and 3, and around lamina S); noradrenergic fibres from the locus coeruleus; serotonergic afferents from the midbrain raphe nuclei; and cholinergic fibres from the pontine and mesencephalic reticular formation.

posterior parts of the temporal association cortex and with the parietal cortex. The medial pulvinar nucleus connects with the inferior parietal cortex, with the posterior cingulate gyrus and with the widespread areas of the temporal lobe, including the posterior parahippocampal gyrus and perirhinal and entorhinal cortex. It also has extensive connections with prefrontal and orbitofrontal cortices. Similarly, the lateral pulvinar nucleus may also connect with the rostromedial prefrontal cortex.

Little is known of the functions of the pulvinar. The inferior pulvinar nucleus contains a complete retinotopic representation, and lateral and medial pulvinar nuclei also contain visually responsive cells. However, the latter nucleus, at least, is not purely visual – other modality responses can be recorded and some cells may be polysensory. Given the complexity of functions of the association areas to which they project, particularly in the temporal lobe (e.g. perception, cognition and memory), it is likely that the role of the pulvinar in modulating these functions is equally complex.

Anteriorly, the major subdivisions of the pulvinar blend into a poorly differentiated region, within which several nuclear components have been recognized, including the anterior or oral pulvinar, the suprageniculatus nucleus and the posterior nuclei. The connectivity of this complex is also not well understood. It is recognized that different components receive subcortical afferents from the spinothalamic tract and the superior and inferior colliculi. Cortical connections centre primarily on the insula and adjacent parts of the parietal operculum posteriorly. Stimulation of this region has been reported to elicit pain, and large lesions may alleviate painful conditions. Similarly, excision of its cortical target in the parietal operculum, or small infarcts in this cortical region, may result in hypoalgesia.

INTRALAMINAR NUCLEI

The term intralaminar nuclei refers to collections of neurones within the internal medullary lamina of the thalamus (see Fig. 23.4). Two groups of nuclei are recognized. The anterior (rostral) group is subdivided into central medial, paracentral and central lateral nuclei. The posterior (caudal) intralaminar group consists of the centromedian and parafascicular nuclei. The designations central medial and centromedian are open to confusion; however, they are an accepted part of the terminology of thalamic nuclei in common usage. The centromedian nucleus is much larger, is considerably expanded in humans in comparison with other species, and is importantly related to the globus pallidus, deep cerebellar nuclei and motor cortex. Anteriorly, the internal medullary lamina separates the mediodorsal nucleus from the ventral lateral complex. It is occupied by the paracentral nucleus laterally, and the central medial nucleus ventromedially, as the two laminae converge towards the midline. A little more posteriorly, the central lateral nucleus appears dorsally in the lamina as the latter splits to enclose the lateral dorsal nucleus. More posteriorly, at the level of the ventral posterior nucleus, the lamina splits to enclose the ovoid centromedian nucleus. The smaller parafascicular nucleus lies more medially.

The anterior intralaminar nuclei, i.e. central medial, paracentral and central lateral, have reciprocal connections with widespread cortical areas. There is some evidence of areal preference: the central lateral nucleus projects mainly to parietal and temporal association areas; the paracentral nucleus to occipitotemporal and prefrontal cortex; and the central medial nucleus to orbitofrontal and prefrontal cortex and to the cortex on the medial surface. In contrast, the posterior nuclei, i.e. centromedian and parafascicular nuclei, have more restricted connections, principally with the motor, premotor and supplementary motor areas. Both anterior and posterior intralaminar nuclei also project to the striatum. Many cells throughout the anterior nuclei have branched axons, which pass to both the cortex and the striatum. Dual projections are less frequent in the posterior nuclei. The thalamostriate projection is topographically organized. The posterior intralaminar nuclei receive a major input from the internal pallidum. Additional afferents come from the pars reticularis of the substantia nigra, the deep cerebellar nuclei, the pedunculopontine nucleus of the midbrain, and possibly the spinothalamic tract. The anterior nuclei have widespread subcortical afferents. The central lateral nucleus receives afferents from the spinothalamic tract, and all component nuclei receive fibres from the brainstem reticular formation, the superior colliculus and several pretectal nuclei. Afferents to all intralaminar nuclei from the brainstem reticular formation include a prominent cholinergic pathway.

The precise functional role of the intralaminar nuclei is unclear. They appear to mediate cortical activation from the brainstem reticular formation and to play a part in sensory–motor integration. Damage to the

intralaminar nuclei may contribute to thalamic neglect, i.e. the unilateral neglect of stimuli originating from the contralateral body or extrapersonal space. This may arise particularly from unilateral damage to the centromedian–parafascicular complex. The latter has been targeted in humans for the neurosurgical control of pain and epilepsy. Bilateral injury to the posterior intralaminar nuclei leads to a kinetic mutism with apathy and loss of motivation. A second syndrome associated with damage involving the intralaminar nuclei is that of unilateral motor neglect, in which there is contralateral paucity of spontaneous movement and motor activity.

MIDLINE NUCLEI

There is considerable divergence between different authors as to which elements of the medial diencephalon constitute the nuclei of the midline thalamic group. In the present account, the midline group of nuclei includes those medial thalamic structures ventral to the central medial nucleus, i.e. the rhomboid and reuniens nuclei, together with the parataenial nuclei more dorsolaterally.

The midline nuclei receive subcortical afferent fibres from the hypothalamus, the periaqueductal grey matter of the midbrain, the spinothalamic tract and the medullary and pontine reticular formation. They are the major thalamic target of ascending noradrenergic and serotonergic axons from the locus coeruleus and raphe nuclei respectively, and they also receive a cholinergic input from the midbrain. Efferents from the midline nuclei pass to the hippocampal formation, the amygdala and the nucleus accumbens. Additional thalamocortical axons reach the cingulate, and possibly orbitofrontal, cortex. The dual cortical and basal nuclear relationship of these nuclei has often led to their being considered a part of the intralaminar system. The cortical projections are reciprocal. The relationships of the midline nuclei clearly identify them as part of the limbic system. There is some evidence that they may play a role in memory and arousal, and, pathologically, may be important in the regulation of seizure activity.

RETICULAR NUCLEUS

The reticular nucleus is a curved lamella of large, deeply staining fusiform cells that wraps around the lateral margin of the thalamus, separated from it by the external medullary lamina. Anteriorly, it curves around the rostral pole of the thalamus to lie between it and the prethalamic nuclei, notably the bed nucleus of the stria terminalis. The nucleus is so named because it is criss-crossed by bundles of fibres that, as they pass between thalamus and cortex, produce a reticular appearance.

The nucleus is thought to receive collateral branches of corticothalamic, thalamocortical and probably thalamostriatal and pallidothalamic fibres as they traverse it. It receives an additional, probably cholinergic, afferent pathway from the nucleus cuneiformis of the midbrain. Broadly speaking, the afferents from the cortex and thalamus are topographically arranged. The reticular nucleus contains visual, somatic and auditory regions, each with a crude topographic representation of the sensorium concerned. Cells within these regions respond to visual, somatic or auditory stimuli with a latency, suggesting that these properties arise from activation by thalamocortical axon collaterals. Only in areas where representations abut do cells show modality convergence.

The efferent fibres from the reticular nucleus pass into the body of the thalamus and are GABAergic. The projections into the main thalamic nuclei broadly, but not entirely, reciprocate the thalamoreticular connections. There may also be projections to the contralateral dorsal thalamus. The reticular nucleus is believed to function in gating information relayed through the thalamus.

HYPOTHALAMUS

The hypothalamus consists of only 4 cm³ of neural tissue, or 0.3% of the total brain. Nevertheless, it contains the integrative systems that, via the autonomic and endocrine effector systems, control fluid and electrolyte balance, food ingestion and energy balance, reproduction, thermoregulation, and immune and many emotional responses.

The hypothalamus extends from the lamina terminalis to a vertical plane posterior to the mammillary bodies, and from the hypothalamic sulcus to the base of the brain beneath the third ventricle. It lies beneath the thalamus (see Fig. 16.6) and anterior to the tegmental part of the subthalamus and the mesencephalic tegmentum. Laterally, it is

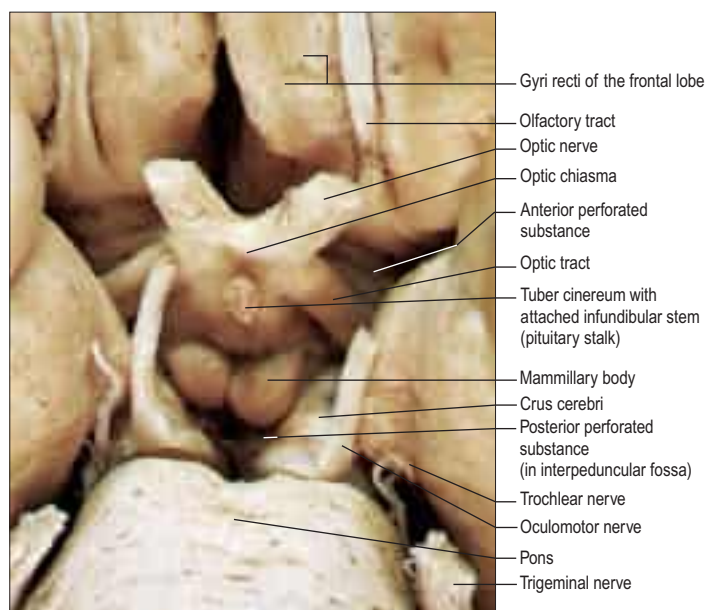


Fig. 23.7 The inferior aspect of the hypothalamus and surrounding structures. The cut end of the trochlear nerve on the left side has folded during processing. (Photograph by Kevin Fitzpatrick, London.)

bordered by the anterior part of the subthalamus, internal capsule and optic tract. Structures in the floor of the third ventricle reach the pial surface in the interpeduncular fossa. From anterior to posterior they are the optic chiasma, the tuber cinereum, the median eminence and the infundibular stalk, the mammillary bodies and the posterior perforated substance (Fig. 23.7). The latter lies in the interval between the diverging crura cerebri, and is pierced by small central branches of the posterior cerebral arteries. It contains the small interpeduncular nucleus, which receives terminals of the fasciculus retroflexus (habenulo-interpeduncular tract) of both sides, and has other connections with the mesencephalic reticular formation and mammillary bodies.

The mammillary bodies are smooth, hemispherical, pea-sized eminences, lying side by side, anterior to the posterior perforated substance, each with nuclei enclosed in fascicles of fibres derived largely from the fornix. The tuber cinereum, between the mammillary bodies and the optic chiasma, is a convex mass of grey matter. From it, the median, conical, hollow infundibulum becomes continuous ventrally with the posterior lobe of the pituitary. Around the base of the infundibulum is the median eminence, which is demarcated by a shallow tubero-infundibular sulcus.

Hypothalamic lesions have long been linked with widespread and bizarre endocrine syndromes and with metabolic, visceral, motor and emotional disturbances. The hypothalamus has major interactions with the neuroendocrine system and the autonomic nervous system, integrating responses to both internal and external afferent stimuli with the complex analysis of the world provided by the cerebral cortex.

The hypothalamus controls the endocrine system in a variety of ways: through magnocellular neurosecretory projections to the posterior pituitary; through parvocellular neurosecretory projections to the median eminence (these control the endocrine output of the anterior pituitary and thereby the peripheral endocrine organs); and via the autonomic nervous system (Nieuwenhuys 1985). The posterior pituitary neurohormones, vasopressin and oxytocin, are primarily involved in the control of osmotic homeostasis and various aspects of reproductive function, respectively. Through its effects on the anterior pituitary, the hypothalamus influences the thyroid gland (thyroid-stimulating hormone, TSH), suprarenal cortex (adrenocorticotrophic hormone, ACTH), gonads (luteinizing hormone, LH; follicle-stimulating hormone, FSH; prolactin), mammary gland (prolactin), and the processes of growth and metabolic homeostasis (growth hormone, GH).

The hypothalamus influences both parasympathetic and sympathetic divisions of the autonomic nervous system. In general, parasympathetic effects predominate when the anterior hypothalamus is stimulated; sympathetic effects depend more on the posterior hypothalamus.

Stimulation of the anterior hypothalamus and paraventricular nucleus can cause decreased blood pressure and decreased heart rate. Stimulation in the anterior hypothalamus induces sweating and vasodilation (and thus heat loss) via projections that pass through the medial forebrain bundle to autonomic centres in the brainstem

and spinal cord. Damage to the anterior hypothalamus, e.g. during surgery for suprasellar extensions of pituitary tumours, can result in an uncontrollable rise in body temperature. Projections to the ventromedial hypothalamus conjointly regulate food intake; damage to these structures during surgery for hypothalamic lesions (e.g. craniopharyngioma) can result in uncontrolled eating and consequent obesity. Stimulation in the posterior part of the hypothalamus induces sympathetic arousal, which produces vasoconstriction, piloerection, shivering and increased metabolic heat production. Circuitry mediating shivering is located in the dorsomedial posterior hypothalamus. However, this does not imply the existence of discrete parasympathetic and sympathetic 'centres'. Stimuli in many different parts of the hypothalamus can cause profound changes in heart rate, cardiac output, vasomotor tone, peripheral resistance, differential blood flow in organs and limbs, the frequency and depth of respiration, motility and secretion in the alimentary tract, erection and ejaculation.

HYPOTHALAMIC NUCLEI

The hypothalamus contains a number of neuronal groups that have been classified on phylogenetic, developmental, cytoarchitectonic, synaptic and histochemical grounds into named nuclei, many of which are not very clearly delimited, especially in the adult. While the hypothalamus contains a few large myelinated tracts, many of the connections are diffuse and unmyelinated, and the precise paths of many afferent, efferent and intrinsic connections are uncertain.

The hypothalamus can be divided anteroposteriorly into chiasmatic (supraoptic), tuberal (infundibulotuberal) and posterior (mammillary) regions, and mediolaterally into periventricular, intermediate (medial) and lateral zones. Between the intermediate and lateral zones is a paramedian plane, which contains the prominent myelinated fibres of the column of the fornix, the mammillothalamic tract and the fasciculus retroflexus. For this reason, some authors group the periventricular and intermediate zones as a single medial zone. These divisions are artificial and functional systems cross them. The main nuclear groups and myelinated tracts are illustrated in Figures 23.8 and 23.9 and are summarized in Table 23.3.

The periventricular zone of the hypothalamus borders the third ventricle. In the anterior wall of the ventricle is the vascular organ of the lamina terminalis (OVLT, organum vasculosum), which is continuous dorsally with the median preoptic nucleus and subfornical organ. On each side in the chiasmatic region are part of the preoptic nucleus, the small, sexually dimorphic suprachiasmatic nucleus and periventricular neurones, which are medial to, and blend with, the paraventricular nucleus. In the tuberal region, the periventricular cell group expands around the base of the third ventricle to form the arcuate nucleus, which overlies the median eminence. In the posterior region, the narrow periventricular zone is continuous laterally with the posterior hypothalamic area and behind that with midbrain periaqueductal grey matter. The periventricular zone also contains a prominent periventricular fibre system.

Suprachiasmatic nucleus

Although it contains only a few thousand neurones, the suprachiasmatic nucleus is a remarkable structure. Sometimes called the master circadian pacemaker in mammals, it appears to be the neural substrate for day–night cycles in motor activity, body temperature, plasma concentration of many hormones, renal secretion, sleeping and waking, and many other variables. Lesions of the suprachiasmatic region lead to a disordered sleep–wake cycle. (For further reading, see Mohawk and Takahashi (2011).)

The suprachiasmatic nucleus has two principal subdivisions: ventrolateral and dorsomedial. Retinal fibres terminate in the ventrolateral subdivision, characterized by neurones immunoreactive for vasoactive intestinal polypeptide (VIP). This appears to be a general input zone, which also receives afferents from the midbrain raphe and parts of the lateral geniculate nucleus of the thalamus. The dorsomedial subdivision has relatively sparse afferent innervation, and characteristically contains parvocellular neurones immunoreactive for arginine vasopressin (AVP). Neurones within the suprachiasmatic nuclei that receive direct retinal input do not respond to pattern, movement or colour. Instead, they operate as luminance detectors, responding to the onset and offset of light, and their firing rates vary in proportion to light intensity, thereby synchronizing to the light–dark cycle.

The suprachiasmatic nucleus receives glutamatergic afferents from retinal ganglion cells that entrain the rhythm to the light–dark cycle, but these are not essential for the production of the rhythm, which

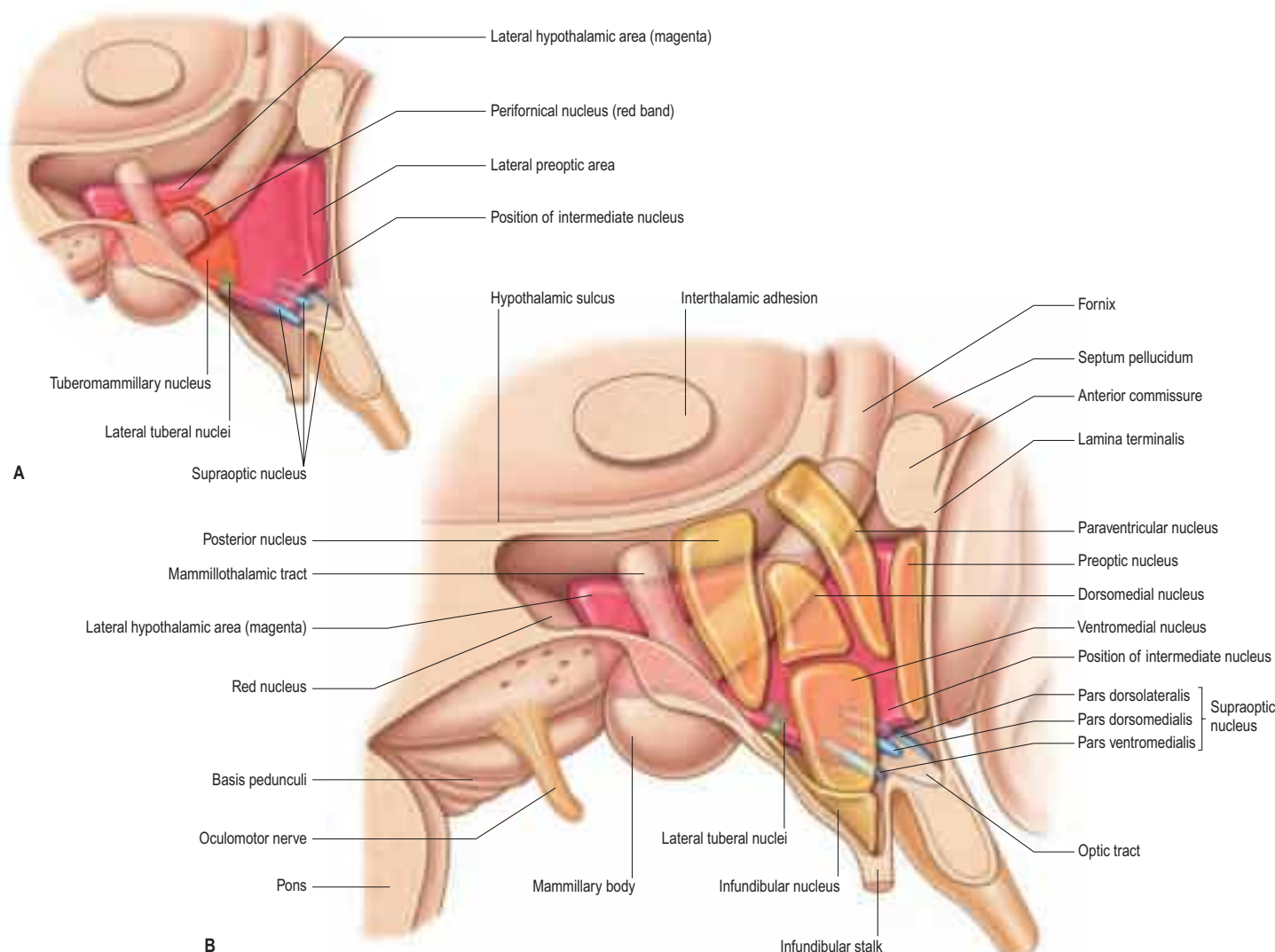


Fig. 23.8 The hypothalamic region of the left cerebral hemisphere viewed from the medial aspect and dissected to display the major hypothalamic nuclei. **A**, The medially placed nuclear groups have been removed. **B**, Both lateral and medial groups are included. Lateral to the fornix and the mamillothalamic tract is the lateral hypothalamic region, in which the tuberomammillary nucleus is situated posteriorly and the lateral preoptic nucleus rostrally. Surrounding the fornix is the perifornical nucleus, which joins the lateral hypothalamic area with the posterior hypothalamic nucleus. The medially placed nuclei (yellow) fill in much of the region between the mamillothalamic tract and the lamina terminalis, but also project caudal to the tract. The lateral tuberal nuclei are situated ventrally, largely in the lateral hypothalamic area. The supraoptic nucleus may form three rather separate parts. The intermediate nuclei form three groups between the supraoptic and paraventricular nuclei. (Modified from Nauta WJH, Haymaker W 1969 *Hypothalamic nuclei and fibre connections*. In: Haymaker W, Anderson E, Nauta WJH (eds) *The Hypothalamus*, by permission of Charles C Thomas Publisher, Ltd, Springfield, Illinois.)

persists in the blind. It contains many different neurotransmitters, including vasopressin, VIP, neuropeptide Y (NPY) and neurotensin. Axons from the suprachiasmatic nucleus pass to many other hypothalamic nuclei, including the paraventricular, ventromedial, dorsomedial and arcuate nuclei.

Neurones producing growth hormone-releasing hormone (GHRH) are largely restricted to the arcuate nucleus. Some extend dorsally into the periventricular nucleus and laterally into the retrochiasmatic area. Their fibres run through the periventricular region to the neurovascular zone of the median eminence. The neurones receive afferent information from glucose receptors in the ventromedial nucleus. Inputs from the hippocampal–amygdala–septal complex could explain the release of GH during stress. In humans, midline defects such as septo-optic dysplasia are associated with defective GH secretion. Dopamine has a stimulatory effect.

Neurones producing somatostatin (growth hormone release-inhibiting hormone) are located in the periventricular nucleus. GHRH and somatostatin are secreted in intermittent (3–5-hour) reciprocal pulses but the origin of the pulses is unclear. A large pulse of GH is secreted at the onset of slow-wave sleep. Somatostatin also inhibits release of pituitary TSH.

Neurones producing gonadotrophin-releasing hormone (GnRH) and projecting to the median eminence are located in the periventricular and arcuate nuclei. Other GnRH neurones are found in the median preoptic area, but these appear to project to the vascular organ of the

lamina terminalis. LH and FSH are secreted by the adenohypophysis in circrhal (hourly) pulses, which are stimulated by GnRH, and are influenced by central monoamine and γ -aminobutyric acid (GABA), by oestrogen and progesterone acting indirectly through other neurones, by corticotrophin-releasing factor, and by endogenous opioids. Gonadotrophin-inhibitory hormone (GnIH) is produced by neurones located in the dorsomedial hypothalamic area that project to the median eminence. GnIH inhibits gonadotrophin synthesis and release by decreasing the activity of GnRH neurones and also by acting directly on pituitary gonadotropes.

Corticotrophin-releasing hormone (CRH) is located primarily in parvocellular periventricular neurones. They are profoundly stimulated by neurogenic (limbic input) and hypoglycaemic (ventromedial nucleus) stress, and are also controlled by negative feedback by cortisol.

Thyrotrophin-releasing hormone (TRH) neurones are rather more widely distributed in the periventricular, ventromedial and dorsomedial nuclei. TRH release is influenced by core temperature, sensed in the anterior hypothalamus, and by negative feedback of thyroid hormones. It stimulates release of pituitary TSH and also acts to excite cold-sensitive neurones in the preoptic area and to inhibit warm-sensitive ones.

Other tubero-infundibular arcuate neurones contain NPY and neurotensin. Arcuate neurones containing pro-opiomelanocortin peptides project to the periventricular nucleus rather than the median eminence.

The suprachiasmatic nucleus also influences the activity of upper spinal preganglionic sympathetic neurones. These neurones in turn project to superior cervical ganglion neurones, which project to the pineal gland. Circadian variation in this postganglionic sympathetic input causes parallel variation in pineal *N*-acetyltransferase activity and thus pineal melatonin production. The role of the pineal gland in humans is uncertain: pineal tumours can influence reproductive development, and administration of melatonin has been advocated to alleviate jet-lag.

Parvocellular neurosecretory neurones lie within the periventricular zone: in particular, in the medial parvocellular part of the paraventricular nucleus, and the arcuate nucleus. The arcuate nucleus is median in the post-infundibular part of the tuber cinereum; it extends forwards into the median eminence and almost encircles the infundibular base, but does not meet anteriorly, where the infundibulum adjoins the median part of the optic chiasma. Its numerous neurones are all small and round in coronal section, and oval or fusiform in sagittal section. No glial layer intervenes between the nucleus and the ependymal tanycytes lining the infundibular recess of the third ventricle. Circadian variation in the secretion of all anterior pituitary hormones suggests that projections from the suprachiasmatic nucleus must reach parvocellular neurosecretory neurones. Afferents from the limbic system probably mediate the widespread effects of stress, and 5-HT (5-hydroxytryptamine, serotonin) and noradrenaline (norepinephrine) from the brainstem influence the output of most anterior pituitary hormones. The axons of parvocellular neurones converge on the infundibulum, forming a tubero-infundibular tract, which ends on the capillary loops that form the hypophyseal portal vessels.

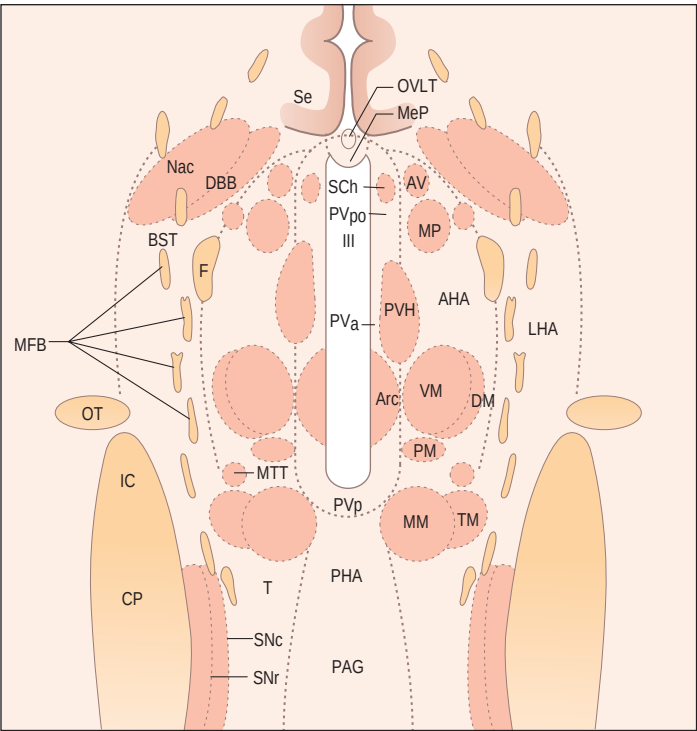


Fig. 23.9 A horizontal section showing the major cell groups and tracts in and around the hypothalamus. Abbreviations: III, third ventricle; AHA, anterior hypothalamic area; Arc, arcuate nucleus; AV, anteroventral preoptic nucleus; BST, bed nucleus of stria terminalis; DBB, nucleus of diagonal band; DM, dorsomedial nucleus; LHA, lateral hypothalamic area; MeP, median preoptic nucleus; MM, mammillary body (mainly medial mammillary nucleus); MP, medial preoptic nucleus; Nac, nucleus accumbens; OVLT, vascular organ of the lamina terminalis; PAG, periaqueductal grey matter; PHA, posterior hypothalamic area; PM, posteromedial nucleus in the chiasmatic/tuberal region of the hypothalamus; PV, periventricular nucleus (PVa, anterior part; PVp, posterior part; PVpo, preoptic part); PVH, paraventricular (hypothalamic) nucleus; SCh, suprachiasmatic nucleus; Se, septal cortex; SNc, substantia nigra pars compacta; SNr, substantia nigra pars reticularis; T, midbrain tegmentum; TM, tuberomammillary nucleus; VM, ventromedial nucleus. Fibre tracts: CP, cerebral peduncle; F, fornix; IC, internal capsule; MFB, medial forebrain bundle; MTT, mammillothalamic tract; OT, optic tract. (Modified with permission from Elsevier. *Progress in Brain Research*, vol 87, Swanson LW, *Biochemical switching in hypothalamic circuits mediating responses to stress*, pp. 181–200, 1991.)

Table 23.3 Main hypothalamic nuclei and their function

Region		Nucleus	Hormone secreted	Function
Chiasmatic	Medial	Median preoptic	GnRH	'Master' clock
		Periventricular	GnRH, somatostatin, CRH, TRH	
	Intermediate	Suprachiasmatic		
Tuberal	Medial	Supraoptic	Vasopressin	Satiety centre
		Paraventricular	Vasopressin, oxytocin	
		Dorsomedial	TRH, ANP	
		Ventromedial	TRH	
Posterior		Arcuate	GHRH, dopamine	Secrete to portal system
	Lateral	Lateral		Feeding centre
	Medial	Mammillary body		Memory

Abbreviations: ANP, atrial natriuretic peptide; CRH, corticotrophin-releasing hormone; GHRH, growth hormone-releasing hormone; GnRH, gonadotrophin-releasing hormone; TRH, thyrotrophin-releasing hormone.

In addition to these peptide-containing cells, dopamine neurones in the arcuate nucleus (A12 group) have terminals in the median eminence and infundibulum. Dopamine acts as the principal prolactin release-inhibiting hormone, and also inhibits secretion of TSH (likewise, TSH acts as a prolactin-releasing hormone). Noradrenergic terminals are found in the median eminence, where they may act largely in a paracrine manner.

The intermediate zone of the hypothalamus contains the best-differentiated nuclei. These are: the paraventricular and supraoptic nuclei; 'intermediate' nuclear groups, which show sexual dimorphism;

ventromedial and dorsomedial nuclei; the mammillary body; and tuberomammillary nuclei. Magnocellular neurosecretory neurones are found in the supraoptic nucleus and paraventricular nucleus, and as isolated clusters of cells between them.

Supraoptic and paraventricular nuclei

The supraoptic nucleus, curved over the lateral part of the optic chiasma, contains a uniform population of large neurones. Behind the chiasma, a thin plate of cells in the floor of the brain forms the retrochiasmatic part.

Supraoptic neurones synthesize vasopressin and they all appear to project to the neurohypophysis. The magnocellular vasopressin neurones detect as little as 1% increase in the osmotic pressure of the blood and stimulate release of vasopressin from the posterior pituitary. A fall in blood volume or blood pressure of greater than 5–10% stimulates the release of vasopressin and the urge to drink. These responses are mediated via volume receptors in the walls of the great veins and atria and baroreceptors in the carotid sinus, which all project via the vagus and glossopharyngeal nerves to the nucleus tractus solitarius and thence to the magnocellular nuclei. A biochemical defect in vasopressin production, or interruption of the supraopticohypophysial pathway (e.g. due to a head injury), can cause cranial diabetes insipidus.

The paraventricular nucleus extends from the hypothalamic sulcus downwards across the medial aspect of the column of the fornix, its ventrolateral angle reaching towards the supraoptic nucleus. Its neurones are more diverse. Magnocellular neurones, which project to the neurohypophysis, tend to lie laterally; parvocellular neurones, which project to the median eminence and infundibulum, lie more medially; and intermediate-sized neurones, which may project caudally, lie posteriorly. The axons of the paraventricular magnocellular neurones pass towards the supraoptic nucleus (paraventriculohypophysial tract), where they join axons of supraoptic neurones to form a supraopticohypophysial tract. This runs down the infundibulum, superficially, and into the neural lobe, where the axons are distended and branch repeatedly around the capillaries. Vasopressin and oxytocin are produced by separate neurones; vasopressin neurones tend to cluster in the ventrolateral part of the paraventricular nucleus and the oxytocin neurones lie around them.

The hypothalamus is essential for the control of pituitary oxytocin, gonadotrophin and prolactin secretion. The release of oxytocin from neurosecretory nerve terminals in the neurohypophysis induces contraction of the uterus and of the myoepithelial cells that surround the mammary gland alveoli. Two neuroendocrine reflexes are involved. Stretching of the cervix of the uterus during childbirth stimulates a multisynaptic afferent pathway that passes via the pelvic plexus, anterolateral column and brainstem to the magnocellular oxytocin neurones (the Ferguson reflex). This is a positive feedback mechanism, which is terminated by the birth of the child. The milk ejection reflex is evoked by suckling of the nipple, which activates intercostal nerves and, thereby, an afferent connection to the hypothalamus. The reflex can be both conditioned to a baby's cry and inhibited by stress.

At the tuberal level, the ventromedial nucleus is well defined by a surrounding neurone-poor zone but the dorsomedial nucleus above it is much less distinct. The ventromedial nucleus contains neurones receptive to plasma levels of glucose and other nutrients, and receives visceral somatic afferents via the nucleus tractus solitarius. The lateral hypothalamus receives olfactory afferents, which act as important food signals. Both areas receive extensive inputs from limbic structures. Stimulation and lesion experiments, together with human case studies, suggest that the ventromedial nuclei act together as a 'satiety centre'. Bilateral ventromedial nucleus damage promotes overeating (hyperphagia), and restricting food intake may provoke rage-like outbursts. The resultant obesity is usually coupled with hyposexuality (Fröhlich syndrome). Interestingly, in infants, ventromedial damage can lead to emaciation despite apparently normal feeding. Experimental lesions in the lateral hypothalamus promote hypophagia or aphagia, while stimulation can prolong feeding, supporting the concept of a lateral hypothalamic 'feeding centre'.

The ventromedial nucleus, lateral hypothalamic area and paraventricular nucleus also influence intermediate metabolism through the autonomic and endocrine systems. These appear to complement the effects on feeding behaviour. Thus, ventromedial stimulation facilitates glucagon release and increases glycogenolysis, gluconeogenesis and lipolysis, whereas lateral hypothalamic stimulation causes insulin release and the opposite metabolic effects. Lesions of the ventromedial nucleus also cause increased vagal and decreased sympathetic tone.

The medial mammillary nuclei, which form the bulk of the mammillary bodies, are very prominent. The distinct existence of a lateral mammillary nucleus is controversial, though a group of larger cells does occur along the lateral border of the medial mammillary nucleus. Lateral to this is the tuberomammillary nucleus, which gives rise to widespread axons that diffusely innervate the entire cerebral cortex, hypothalamus and brainstem.

The lateral zone of the hypothalamus forms a continuum that runs from the preoptic nucleus through the lateral hypothalamic area to the posterior hypothalamus. In the tuberal region, the lateral tuberal nuclei are large and well defined, and surrounded by fine fibres.

CONNECTIONS OF THE HYPOTHALAMUS

The hypothalamus has reciprocal interaction with the rest of the body via two (possibly three) quite distinct routes: neural connections, the bloodstream and (probably) the cerebrospinal fluid.

Some hypothalamic neurones have specific receptors that sense the temperature, osmolarity, glucose, free fatty acid and hormone content of the blood. Neurosecretory neurones secrete neurohormones into the blood. These control the anterior pituitary and act on organs such as the kidney, breast, uterus and blood vessels. Some of the neural connections, especially those to the mammillary bodies, form discrete myelinated fascicles, but most are diffuse and unmyelinated, and their origin and termination are uncertain. Most pathways are multisynaptic and many of the synapses on hypothalamic neurones are derived from intrinsic interneurones.

Broadly, neural inputs to the hypothalamus are derived from the ascending visceral and somatic sensory systems, the visual and olfactory systems, and numerous tracts from the brainstem, thalamus, 'limbic' structures and neocortex. Efferent neural projections are reciprocal to most of these and, in particular, they impinge on and control the central origins of autonomic nerve fibres. The hypothalamus therefore exerts control via the autonomic and endocrine systems and through its connections to the telencephalon.

Afferent connections

The hypothalamus receives visceral, gustatory and somatic sensory information from the spinal cord and brainstem. It receives largely polysynaptic projections from the nucleus tractus solitarius, probably directly and indirectly via the parabrachial nucleus and medullary noradrenergic cell groups (ventral noradrenergic bundle); collaterals of lemniscal somatic afferents (to the lateral hypothalamus); and projections from the dorsal longitudinal reticular formation. Many enter via the medial forebrain bundle (see Fig. 23.9) and periventricular fibre system. Others converge in the midbrain tegmentum, forming the mammillary peduncle to the mammillary body.

The major forebrain inputs to the hypothalamus are derived from structures in the limbic system, including the hippocampal formation, amygdala and septum, and from the piriform lobe and adjacent neocortex. These connections, which are reciprocal, form prominent fibre systems, i.e. the fornix, stria terminalis and ventral amygdalofugal tracts.

The hippocampal formation – in particular, the subiculum and CA1 – is reciprocally connected to the hypothalamus by the fornix, a complex tract that also contains commissural connections. As the fornix curves ventrally towards the anterior commissure, it is joined by fascicles from the cingulate gyrus, indusium griseum and the septal area. It divides around the anterior commissure into pre- and postcommissural parts. The precommissural fornix is distributed to the septum and preoptic hypothalamus, and the septum in turn sends numerous fibres to the hypothalamus. The postcommissural fornix passes ventrally and posteriorly through the hypothalamus to the medial mammillary nucleus. In its course it gives off many fibres to the medial and lateral hypothalamic nuclei.

The amygdala innervates most hypothalamic nuclei anterior to the mammillary bodies. Its corticomедial nucleus innervates preoptic and anterior hypothalamic areas and the ventromedial nucleus. The central nuclei project to the lateral hypothalamus. The fibres reach the hypothalamus by two routes. The short ventral amygdalofugal path passes medially over the optic tract, beneath the lentiform complex, to reach the hypothalamus. The long curved stria terminalis runs parallel to the fornix, separated from it by the lateral ventricle, passes through the bed nucleus of the stria terminalis, and is then distributed to the anterior hypothalamus via the medial forebrain bundle.

Olfactory afferents reach the hypothalamus largely via the nucleus accumbens and septal nuclei, and most terminate in the lateral hypothalamus.

Visual afferents leave the optic chiasma and pass dorsally into the suprachiasmatic nucleus. No auditory connections have been identified, though it is clear that such stimuli influence hypothalamic activity. However, many hypothalamic neurones respond best to complex sensory stimuli, suggesting that sensory information reaching the neocortex has converged and been processed by the amygdala, hippocampus and neocortex. Neocortical corticohypothalamic afferents to the hypothalamus are poorly defined, but probably arise from frontal and insular cortices. Some may relay in the mediodorsal thalamic nucleus and project into the hypothalamus via the periventricular route. Other direct corticohypothalamic fibres may end in lateral, dorsomedial, mammillary and posterior hypothalamic nuclei, but all these connections are questioned.

Like the rest of the forebrain, the hypothalamus also receives diffuse aminergic inputs from the locus coeruleus and raphe nuclei. In addition, it receives input from the ventral tegmental ascending cholinergic pathway: a noradrenergic input to dorsomedial, periventricular, paraventricular, supraoptic and lateral hypothalamic nuclei from the ventral tegmental noradrenergic bundle, and fibres from the mesolimbic dopaminergic system. Group A11 innervates the medial hypothalamic nuclei, and groups A13 and A14 supply the dorsal and rostral hypothalamic nuclei. Many of these fibres also run in the medial forebrain bundle.

The medial forebrain bundle is a loose grouping of fibre pathways that mostly run longitudinally through the lateral hypothalamus (see Fig. 23.9). It connects forebrain autonomic and limbic structures with the hypothalamus and brainstem, receiving and giving small fascicles throughout its course. It contains descending hypothalamic afferents from the septal area and orbitofrontal cortex, ascending afferents from the brainstem, and efferents from the hypothalamus.

Efferent connections

Hypothalamic efferents include reciprocal paths to the limbic system, descending polysynaptic paths to autonomic and somatic motor neurones, and neural and neurovascular links with the pituitary.

Septal areas and the amygdaloid complex have reciprocal hypothalamic connections along the paths described above. The medial preoptic and anterior hypothalamic areas give short projections to nearby hypothalamic groups. The ventromedial nucleus has more extensive projections that pass via the medial forebrain bundle to the bed nucleus of the stria terminalis, nucleus basalis (basal nucleus of Meynert), central nucleus of the amygdala, and midbrain reticular formation. The posterior hypothalamus projects largely to the midbrain central grey matter. Some tuberal and posterior lateral hypothalamic neurones project directly to the entire neocortex and appear to be essential for maintaining cortical arousal, but the topography of these projections is unclear.

Hypothalamic neurones projecting to autonomic neurones are found in the paraventricular nucleus (oxytocin and vasopressin neurones), perifornical and dorsomedial nuclei (atrial natriuretic peptide neurones), lateral hypothalamic area (α -melanocyte-stimulating hormone; α -MSH neurones) and zona incerta (dopamine neurones). These fibres run through the medial forebrain bundle into the tegmentum, ventrolateral medulla and dorsal lateral funiculus of the spinal cord. In the brainstem, fibres innervate the parabrachial nucleus, nucleus ambiguus, nucleus of the solitary tract and dorsal motor nucleus of the vagus. In the spinal cord, they end on sympathetic and parasympathetic preganglionic neurones in the intermediolateral column. Both oxytocin- and vasopressin-containing fibres can be traced to the most caudal spinal autonomic neurones.

The medial mammillary nucleus gives rise to a large ascending fibre bundle, which diverges into the mammillothalamic and mammillotegmental tracts (see Fig. 23.9). The mammillothalamic tract ascends through the lateral hypothalamus to reach the anterior thalamic nuclei, whence massive projections radiate to the cingulate gyrus. The mammillotegmental tract curves inferiorly into the midbrain ventral to the medial longitudinal fasciculus, and is distributed to tectal reticular nuclei.

PITUITARY GLAND

The pituitary gland, or hypophysis cerebri (Fig. 23.10; see Fig. 18.10), is a reddish-grey, ovoid body, about 12 mm in transverse and 8 mm in anteroposterior diameter, and with an average adult weight of 500 mg. It is continuous with the infundibulum, a hollow, conical, inferior process from the tuber cinereum of the hypothalamus. It lies within the pituitary fossa of the sphenoid bone, where it is covered superiorly by



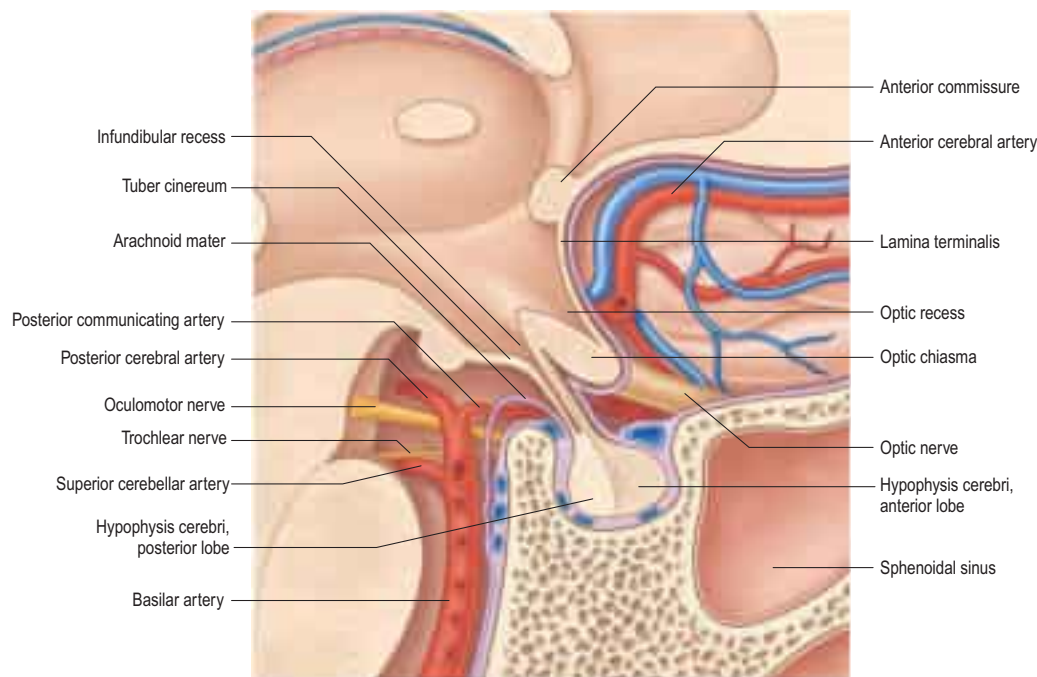


Fig. 23.10 A median section through the hypophysis cerebri.

a circular diaphragma sellae of dura mater. The latter is pierced centrally by an aperture for the infundibulum and separates the anterior superior aspect of the pituitary from the optic chiasma. Inferiorly, the pituitary is separated from the floor of the fossa by a venous sinus that communicates with the circular sinus. The meninges blend with the pituitary capsule and are not separate layers.

The pituitary (Sam and Frohman 2008) has two major parts, the neurohypophysis and adenohypophysis, which differ in their origin, structure and function. The neurohypophysis is a diencephalic down-growth connected with the hypothalamus. The adenohypophysis is an ectodermal derivative of the stomatodeum. Both include parts of the infundibulum (whereas the older terms 'anterior lobe' and 'posterior lobe' do not). The infundibulum has a central infundibular stem, which contains neural hypophysial connections and is continuous with the median eminence of the tuber cinereum. Thus, the term neurohypophysis includes the median eminence, infundibular stem and neural lobe or pars posterior. Surrounding the infundibular stem is the pars tuberalis, a component of the adenohypophysis. The main mass of the adenohypophysis may be divided into the pars anterior (pars distalis) and the pars intermedia, which are separated in fetal and early postnatal life by the hypophysial cleft, a vestige of Rathke's pouch, from which it develops.

The neurohypophysis includes the pars posterior (pars nervosa, posterior or neural lobe), infundibular stem and median eminence. The adenohypophysis includes the pars anterior (pars distalis or glandularis), pars intermedia and pars tuberalis.

Neurohypophysis

In early fetal life, the neurohypophysis contains a cavity continuous with the third ventricle. Axons arising from groups of hypothalamic neurones (e.g. the magnocellular neurones of the supraoptic and paraventricular nuclei) terminate in the neurohypophysis. The long magnocellular axons pass to the main mass of the neurohypophysis. They form the neurosecretory hypothalamohypophysial tract and terminate near the sinusoids of the posterior lobe. Some smaller parvocellular neurones in the periventricular zone have shorter axons, and end in the median eminence and infundibular stem among the superior capillary beds of the venous portal circulation. These small neurones produce releasing and inhibitory hormones, which control the secretory activities of the adenohypophysis via its portal blood supply.

The neurohormones stored in the main part of the neurohypophysis are vasopressin (antidiuretic hormone; ADH), which controls reabsorption of water by renal tubules, and oxytocin, which promotes the contraction of uterine smooth muscle in childbirth and the ejection of milk from the breast during lactation (pituitary hormones are summarized in Table 23.4). Storage granules containing active hormone polypeptides bound to a transport glycoprotein, neurophysin, pass down axons from their site of synthesis in the neuronal somata (Fig. 23.11).

Adenohypophysis

The adenohypophysis is highly vascular. It consists of epithelial cells of varying size and shape arranged in cords or irregular follicles, between which lie thin-walled vascular sinusoids supported by a delicate reticular connective tissue (Fig. 23.12). Most of the hormones synthesized by the adenohypophysis are trophic (see Table 23.4 for a summary of pituitary hormones). They include the peptides GH, involved in the control of body growth, and prolactin, which stimulates both growth of breast tissue and milk secretion. Glycoprotein trophic hormones are the large pro-opiomelanocortin precursor of ACTH, which controls the secretion of certain suprarenal cortical hormones; TSH; FSH, which stimulates growth and secretion of oestrogens in ovarian follicles and spermatogenesis (acting on testicular Sertoli cells); and LH, which induces progesterone secretion by the corpus luteum and testosterone synthesis by Leydig cells in the testis. Pro-opiomelanocortin is cleaved into a number of different molecules, including ACTH. Beta-lipotropin is released from the pituitary but its lipolytic function in humans is uncertain. Beta-endorphin is another cleavage product released from the pituitary.

Neurones that secrete the peptides and amines that control the anterior lobe are widely distributed within the hypothalamus. They are situated mainly in the medial zone, in the arcuate nucleus, medial parvocellular part of the paraventricular nucleus, and periventricular nucleus.

Arteries and veins of the pituitary gland

The arteries of the pituitary arise from the internal carotid arteries via a single inferior and several superior hypophysial arteries on each side (Page 1982). The former come from the cavernous part of the internal carotid artery, the latter from its supraclinoid part and from the anterior and posterior cerebral arteries. The inferior hypophysial arteries divide into medial and lateral branches, which anastomose across the midline and form an arterial ring around the infundibulum. Fine branches from this circular anastomosis enter the neurohypophysis to supply its capillary bed. The superior hypophysial arteries supply the median eminence, upper infundibulum and, via the artery of the trabecula, the lower infundibulum. (The trabecula is a compact band of connective tissue and blood vessels lying within the pars distalis on either side of the midline; it forms a prominent fibrovascular tuft close to the junction of the central and lateral parts of the pars distalis; Xuereb et al (1954).) A confluent capillary net, extending through the neurohypophysis, is supplied by both sets of hypophysial vessels. Reversal of flow can occur in cerebral capillary beds lying between the two supplies.

The arteries of the median eminence and infundibulum end in characteristic sprays of capillaries, which are most complex in the upper infundibulum. In the median eminence, these form an external or 'mantle' plexus and an internal or 'deep' plexus. The external plexus, fed

Usually obliterated in childhood, remnants may persist in the form of cystic cavities, often present near the adenoneurohypophysial frontier, which sometimes invade the neural lobe. The human pars intermedia is rudimentary. It may be partially displaced into the neural lobe, and has been included in the anterior and posterior parts by different observers. Apart from this equivocation, which is of little significance, the pars anterior and pars posterior may be equated with the anterior and posterior lobes. When the associated infundibular parts continuous with these lobes are included, the names adenohypophysis and neurohypophysis become appropriate.

The thin, unmyelinated axons of the neurohypophysis are ensheathed by typical astrocytes in the infundibulum (see Fig. 23.12). Near the posterior lobe, astrocytes are replaced by pituicytes. These are dendritic neuroglial cells of variable appearance, often with long processes running parallel to adjacent axons; they constitute most of the non-excitabile tissue in the neurohypophysis. Typically, their cytoplasmic processes end on the walls of capillaries and sinusoids between nerve terminals. Axons also end in perivascular spaces; they lie close to the walls of sinusoids but remain separated from them by two basal laminae, one around the nerve endings and the other underlying the fenestrated endothelial cells. The spaces between the basal laminae are occupied by fine collagen fibrils.

The epithelial endocrine cells, which secrete the different adenohypophysial hormones, may be distinguished in part by their differing affinities for acidic and basic dyes. Cells staining strongly are described as chromophils, while those with low affinity for dyes are chromophobes. Acidophils stain strongly with acidic dyes, whereas basophils, which are more prevalent in the central part of the gland, stain strongly with basic dyes. Cells may also be classified according to the hormones they synthesize into somatotrophs (GH-secreting acidophils, the most numerous chromophil type); lactotrophs (prolactin-secreting acidophils, which are dominant in pregnancy and hypertrophy during lactation); gonadotrophs (FSH- and LH-secreting basophils); thyrotrophs (TH-secreting basophils); and corticotrophs (ACTH-secreting basophils). Chromophobes are thought to be quiescent or degranulated chromophils, or immature precursor cells, and constitute up to half of the cells of the adenohypophysis.

The pars intermedia contains follicles of chromophobe cells that surround cyst-like structures lined by epithelium and are filled, to varying degrees, with glycosylated colloidal material. Secretory products of this region may include cleavage products of pro-opiomelanocortin but their functional significance is uncertain. The pars tuberalis contains a large number of blood vessels, between which are cords or clusters of gonadotrophs and undifferentiated cells.

Table 23.4 Pituitary hormones

Hormone	Cell type	Function / effector site	Releasing factor	Inhibitor
ACTH	Corticotrophs (basophils)	Suprarenal cortex	CRH	
TSH	Thyrotrophs (basophils)	Thyroid gland	TRH	Somatostatin
LH, FSH	Gonadotrophs (basophils)	Gonads	GnRH (pulsatile)	GnIH
GH	Somatotrophs (acidophils)	Body growth and metabolism	GHRH	Somatostatin
Prolactin	Lactotrophs (acidophils)	Mammary gland		Dopamine
Vasopressin (ADH)	Magnocellular hypothalamic neurons	Water absorption in distal renal tubule		
Oxytocin	Magnocellular hypothalamic neurons	Contraction of uterine smooth muscle and ejection of breast milk		

Abbreviations: ACTH, adrenocorticotrophic hormone; ADH, antidiuretic hormone; CRH, corticotrophin-releasing hormone; FSH, follicle-stimulating hormone; GH, growth hormone; GHRH, growth hormone-releasing hormone; GnIH, gonadotrophin-inhibitory hormone; GnRH, gonadotrophin-releasing hormone; LH, luteinizing hormone; TRH, thyrotrophin-releasing hormone; TSH, thyroid-stimulating hormone.

A small collection of adenohypophysial tissue lies in the mucoperiosteum of the human nasopharyngeal roof. By 28 weeks *in utero* it is well vascularized and capable of secretion, receiving blood from the systemic vessels of the nasopharyngeal roof. At this stage, it is covered posteriorly by fibrous tissue. This is replaced in the second half of fetal life by venous sinuses, and a trans-sphenoidal portal venous system develops, bringing the nasopharyngeal tissue under the same hypothalamic control as the cranial adenohypophysial tissue. The peripheral vascularity of the pharyngeal hypophysis persists until about the fifth year. The organ is then reinvested by fibrous tissue and presumed to be controlled once more by factors present in systemic blood. Though it does not change in size after birth in males, in females it becomes smaller, returning to natal volume during the fifth decade, when once again it may be controlled via a trans-sphenoidal extension of the hypothalamohypophysial portal venous system. The human pharyngeal hypophysis may be a reserve of potential adenohypophysial tissue, which may be stimulated, particularly in females, to synthesize and secrete adenohypophysial hormones in middle age, when intracranial adenohypophysial tissue is beginning to fail.

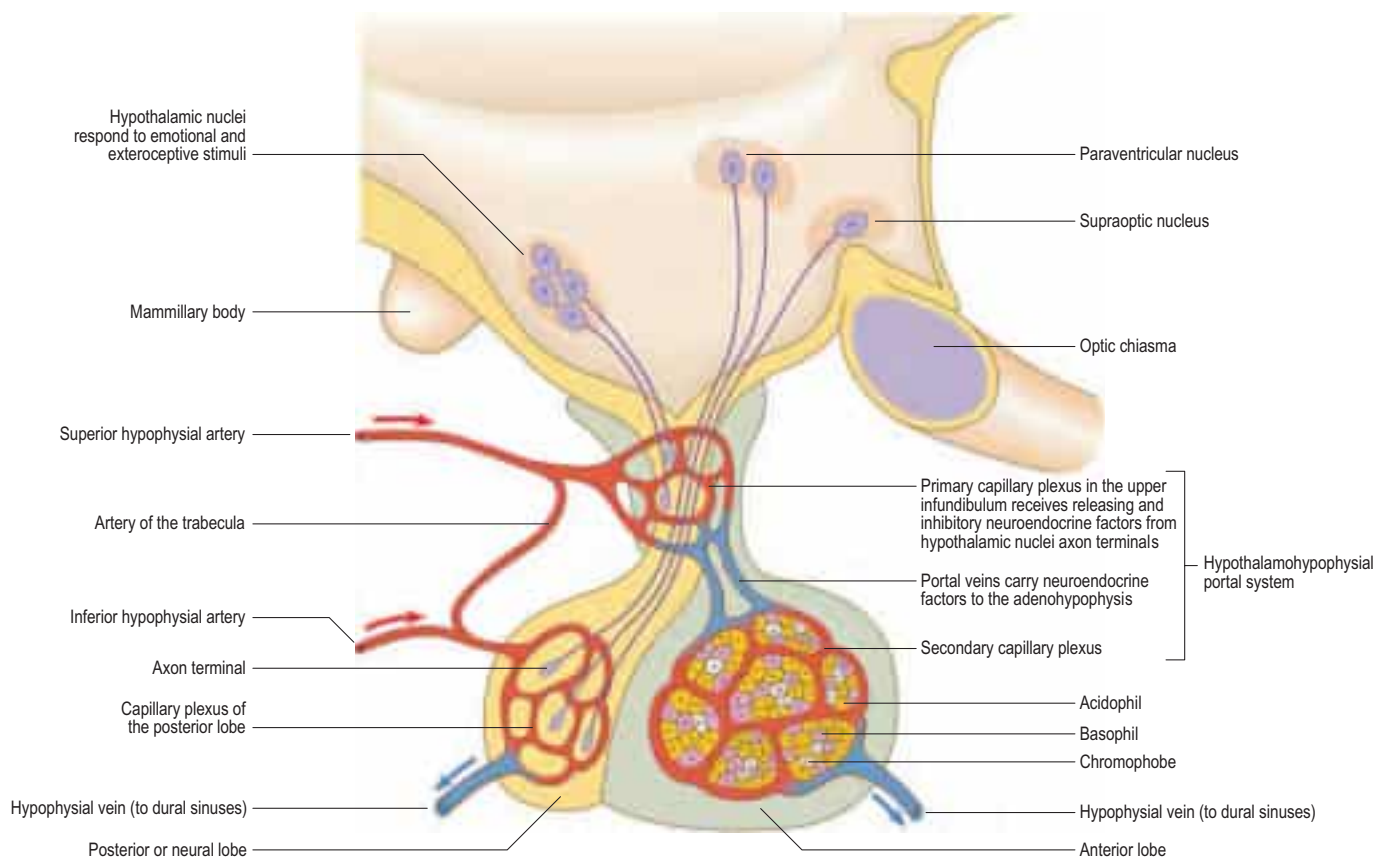


Fig. 23.11 The main systems controlling the endocrine secretory activities of the pituitary gland.

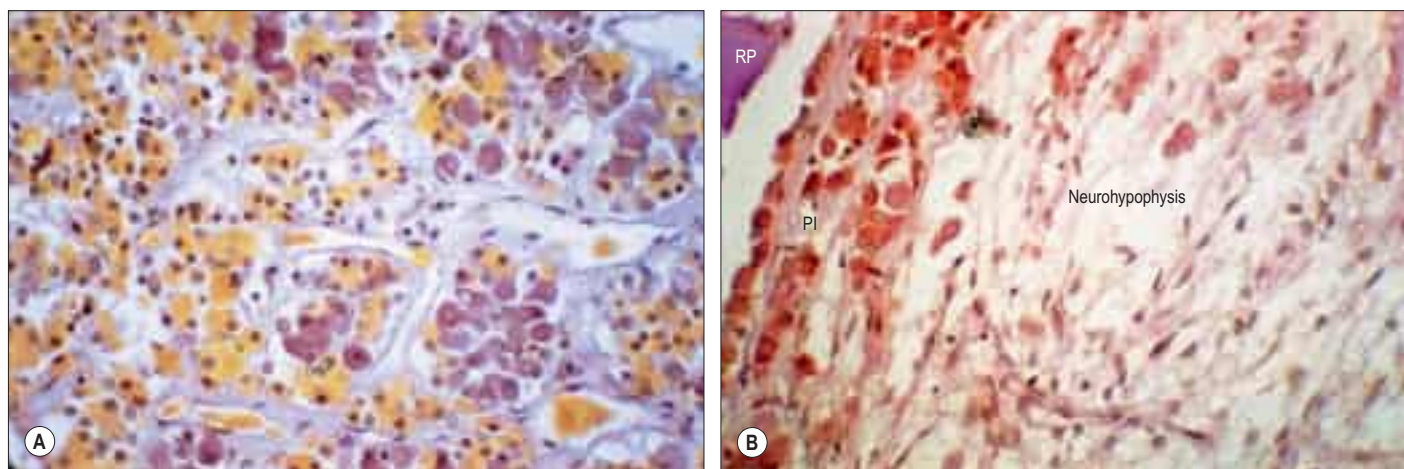


Fig. 23.12 The pituitary gland (trichrome-stained). **A**, The endocrine cells of the adenohypophysis. Chromophils can be distinguished as acidophils (yellow) and basophils (pink). A network of sinusoids is seen between clusters of secretory cells. **B**, The neurohypophysis (right), with nerve fibres and pituicytes. To the left is the pars intermedia (PI) with scattered, deeper-staining secretory cells and a cyst containing colloid (top left), representing the remnants of Rathke's pouch (RP).

by the superior hypophyseal arteries, is continuous with the infundibular plexus and is drained by long portal vessels, which descend to the pars anterior. The internal plexus lies within and is supplied by the external plexus. It is continuous posteriorly with the infundibular capillary bed and, like the external plexus, is drained by long portal vessels. Short portal vessels run from the lower infundibulum to the pars anterior. Both types of portal vessel open into vascular sinusoids, which lie between the secretory cords in the adenohypophysis and provide most of its blood. There is no direct arterial supply. The portal system carries hormone-releasing factors, probably elaborated in parvocellular groups of hypothalamic neurones, and these control the secretory cycles of cells in the pars anterior. The pars intermedia appears to be avascular.

There are three possible routes for venous drainage of the neurohypophysis: to the adenohypophysis, via long and short portal vessels; into the dural venous sinuses, via the large inferior hypophyseal veins; and to the hypothalamus, via capillaries passing to the median eminence. The venous drainage carries hypophyseal hormones from the

gland to their targets and also facilitates feedback control of secretion. However, venous drainage of the adenohypophysis appears restricted: few vessels connect it directly to the systemic veins and so the routes by which blood leaves remain obscure.

SUBTHALAMUS

The subthalamus is a complex region of nuclear groups and fibre tracts (see Fig. 24.12). The main nuclear groups are the subthalamic nucleus, the reticular nucleus, the zona incerta, the fields of Forel and the pre-geniculate nucleus. The rostral poles of the red nucleus and substantia nigra also extend into this area.

The main subthalamic tracts are: the upper parts of the medial, spinal and trigeminal lemnisci and the solitariothalamic tract, all approaching their terminations in the thalamic nuclei; the dento-thalamic tract from the contralateral superior cerebellar peduncle

accompanied by ipsilateral rubrothalamic fibres; the fasciculus retroflexus; the fasciculus lenticularis; the fasciculus subthalamicus; the ansa lenticularis; fascicles from the prerubral field (H field of Forel); the continuation of the fasciculus lenticularis (in the H₂ field of Forel); and the fasciculus thalamicus (the H₁ field of Forel).

SUBTHALAMIC NUCLEUS

The subthalamic nucleus is structurally and functionally related to the basal ganglia and is therefore considered with them on [page 369](#).

ZONA INCERTA AND FIELDS OF FOREL

The zona incerta is an aggregation of small cells that lies between the ventral part of the external medullary lamina of the thalamus and the cerebral peduncle. It is linked to the reticular nucleus dorsolaterally. More medially there is a scattered group of cells in a matrix of fibres known as the H field of Forel (see [Fig. 24.11](#)). Field H₁ of Forel consists of the thalamic fasciculus, which lies dorsal to the zona incerta. Field H₂ of Forel contains the fasciculus lenticularis and lies ventrally, between the zona incerta and the subthalamic nucleus (see [Figs 24.6, 24.12](#)).

The zona incerta receives fibres from the sensorimotor cortex, the pregeniculate nucleus, the deep cerebellar nuclei, the trigeminal nuclear complex and the spinal cord. It projects to the spinal cord and the pretectal region. Its functions are unknown.

The neurones of the H field of Forel receive afferents from the spinal cord, the reticular formation of the brainstem and, possibly, the internal pallidum. They may project to the spinal cord. Like the zona incerta, their functions are unknown.

In addition to terminal parts of the lemniscal, dentothalamic and rubrothalamic tracts, the subthalamus contains massive fibre tracts derived from the globus pallidus. The fasciculus lenticularis is the dorsal component of pallidofugal fibres that traverse the internal capsule. It turns medially near the medial aspect of the capsule, partly intermingled with the dorsal zone of the subthalamic nucleus and the ventral part of the zona incerta, where the fasciculus traverses the H₂ field of Forel. Reaching the medial border of the zona incerta, the fasciculus intermingles with fibres of the ansa lenticularis, scattered elements of the prerubral nucleus, and dentothalamic and rubrothalamic fibres. This merging of diverse pathways and associated cell groups is variously called the prerubral, tegmental or H field of Forel.

The ansa lenticularis has a complex origin from both parts of the globus pallidus and possibly other adjacent structures. It curves medially round the ventral border of the internal capsule, and continues dorsomedially to mingle with other fibres in the prerubral field. Some fibres in the fasciculus lenticularis and ansa lenticularis synapse in the subthalamic nucleus, prerubral field and zona incerta. The remainder continue laterally, with other fascicles, into the thalamic nuclei, particularly the ventral anterior, ventral lateral and centromedian nuclei.

The thalamic fasciculus extends from the prerubral field, and its territory is termed the H₁ field of Forel. It lies dorsal to, and also partly traverses, the zona incerta, and is related dorsally to the ventral thalamic nuclei. It contains continuations of the fasciculus lenticularis and ansa lenticularis, and dentothalamic, rubrothalamic and thalamostriate fibres.

The subthalamic fasciculus connects the subthalamic nucleus with the globus pallidus. It contains an abundant two-way array of fibres that traverse the internal capsule, interweaving with it at right angles.

EPITHALAMUS

The epithalamus consists of the anterior and posterior paraventricular nuclei, the medial and lateral habenular nuclei, the stria medullaris thalami, posterior commissure and the pineal body.

HABENULAR NUCLEI AND STRIA MEDULLARIS THALAMI

The habenular nuclei lie posteriorly at the dorsomedial aspect of the thalamus, immediately deep to the ependyma of the third ventricle, with the stria medullaris thalami above and laterally. The medial habenular nucleus is a densely packed, deeply staining mass of cholinergic neurones, whereas the lateral nucleus is more dispersed and paler-staining. The fasciculus retroflexus (habenulo-interpeduncular tract) emerges from the ventral margin of the nuclei and courses ventrally, skirts the inferior zone of the thalamic mediodorsal nucleus, and

traverses the superomedial region of the red nucleus to reach the interpeduncular nucleus. The habenular nuclear complex is limited laterally by a fibrous lamina, which enters the fasciculus retroflexus. Posteriorly, the nuclei of the two sides and the internal medullary laminae are linked across the midline by the habenular commissure. The tela choroidea of the third ventricle usually arises from the ependyma at the superolateral corner of the medial habenular nucleus.

Afferent fibres to the habenular nuclei travel in the stria medullaris from the prepiriform cortex bilaterally, the nucleus basalis (of Meynert) and the hypothalamus. Afferents from the internal pallidum ascend through the thalamus and may be collaterals of pallidothalamic axons. Additional inputs come from the pars compacta of the substantia nigra, the midbrain raphe nuclei and the lateral dorsal tegmental nucleus. The afferent pathways mostly end in the lateral habenular nucleus. The only identified afferent fibres to the medial habenular nucleus come from the septofimbrial nucleus.

The medial habenular nucleus sends efferent fibres to the interpeduncular nucleus of the midbrain. The lateral habenular nucleus sends fibres to the raphe nuclei and the adjacent reticular formation of the midbrain, the pars compacta of the substantia nigra and the ventral tegmental area, and to the hypothalamus and basal forebrain.

The main habenular outflow reaches the interpeduncular nucleus, mediodorsal thalamic nucleus, mesencephalic tectum and reticular formation, the largest component constituting the fasciculus retroflexus to the interpeduncular nucleus. The latter provides relays to the midbrain reticular formation, from which tectotegmentospinal tracts and dorsal longitudinal fasciculi connect with autonomic preganglionic neurones controlling salivation, gastric and intestinal secretory activity and motility, and motor nuclei for mastication and deglutition.

The stria medullaris crosses the superomedial thalamic aspect, skirts medial to the habenular trigone and sends many fibres into the ipsilateral habenula. Other fibres cross in the anterior pineal lamina and decussate, as the habenular commissure, to reach the contralateral habenula. Some fibres are really commissural and interconnect the amygdaloid complexes and hippocampal cortices. They are accompanied by crossed tectohabenular fibres. Serotonin-containing fibres from the ventral ascending tegmental serotonergic bundle, which join the fasciculus retroflexus to reach the nuclei, may control neurones of the habenulopineal tract, and thus influence innervation of pinealocytes. Similarly, habenular nuclear afferents from the dorsal ascending tegmental noradrenergic bundle may influence pinealocytes.

Little is known of the physiological functions of the habenular nuclei. It has been suggested that they may be involved in the control of sleep mechanisms. Though the human habenula is relatively small, it is a focus of integration of diverse olfactory, visceral and somatic afferent paths. Lesions that include this area of the medial diencephalon indicate that it plays a role in the regulation of visceral and neuroendocrine functions. Ablation of the habenula causes extensive changes in metabolism and in endocrine and thermal regulation.

POSTERIOR COMMISSURE

The posterior commissure, which is of unknown constitution in humans, is a small fasciculus that decussates in the inferior pineal lamina. Various small nuclei are associated with it. Among these are the interstitial nuclei of the posterior commissure, the nucleus of Darkschewitsch in the periaqueductal grey matter, and the interstitial nucleus of Cajal near the upper end of the oculomotor complex, closely linked with the medial longitudinal fasciculus. Fibres from all these nuclei and the fasciculus cross in the posterior commissure. It also contains fibres from thalamic and pretectal nuclei and the superior colliculi, together with fibres that connect the tectal and habenular nuclei. The destinations and functions of many of these fibres are obscure.

PINEAL GLAND

The pineal gland or epiphysis cerebri (see [Figs 18.10, 25.48B](#)) is a small, reddish-grey organ, occupying a depression between the superior colliculi. It is inferior to the splenium of the corpus callosum, from which it is separated by the tela choroidea of the third ventricle and the contained cerebral veins. It is enveloped by the lower layer of the tela, which is reflected from the gland to the tectum. The pineal is about 8 mm long. Its base, directed anteriorly, is attached by a peduncle, which divides into inferior and superior laminae, separated by the pineal recess of the third ventricle, and containing the posterior and habenular commissures respectively. Aberrant commissural fibres may invade the gland but do not terminate near parenchymal cells.

Septa extend into the pineal gland from the surrounding pia mater. They divide the gland into lobules and carry blood vessels and fine unmyelinated sympathetic axons. The gland has a rich blood supply. The pineal arteries are branches of the medial posterior choroidal arteries, which are themselves branches of the posterior cerebral artery. Within the gland, branches of the arteries supply fenestrated capillaries whose endothelial cells rest on a tenuous and sometimes incomplete basal lamina. The capillaries drain into numerous pineal veins, which open into the internal cerebral veins and/or into the great cerebral vein.

Postganglionic adrenergic sympathetic axons (derived from neurones in the superior cervical ganglion) enter the dorsolateral aspect of the gland from the region of the tentorium cerebelli as the nervus conarii, which may be single or paired. The nerve lies deep to the endothelium of the wall of the straight sinus and is associated with blood vessels and parenchymal cells within the pineal.

The pineal gland contains cords and clusters of pinealocytes, associated with astrocyte-like neuroglia. Neuroglia are the main cellular component of the pineal stalk. Pinealocytes are highly modified neurones. They contain multiple synaptic ribbons, randomly distributed between adjacent cells, and are coupled by gap junctions. Two or more processes extend from each cell body and end in bulbous expansions near capillaries or, less frequently, on ependymal cells of the pineal recess. These terminal expansions contain rough endoplasmic reticulum, mitochondria and dense-core vesicles that store melatonin. Melatonin, and its precursor 5-HT (serotonin), are synthesized from tryptophan by the pinealocytes, and secreted into the surrounding network of fenestrated capillaries.

The pineal is an endocrine gland of major regulatory importance (Macchi and Bruce 2004). It modifies the activity of the adenohypophysis, neurohypophysis, endocrine pancreas, parathyroids, suprarenal cortex, suprarenal medulla and gonads. Its effects are largely inhibitory. Indoleamine and polypeptide hormones secreted by pinealocytes are believed to reduce the synthesis and release of hormones by the pars anterior, either by direct action on its secretory cells or indirectly by inhibiting production of hypothalamic releasing factors. Pineal secretions may reach their target cells via the cerebrospinal fluid or the blood stream. Some pineal indoleamines, including melatonin and enzymes for their biosynthesis (e.g. 5-HT, *N*-acetyltransferase), show circadian rhythms in concentration. The level rises during darkness, and falls during the day, when secretion may be inhibited by sympathetic activity. It is thought that the intrinsic rhythmicity of an endogenous circadian

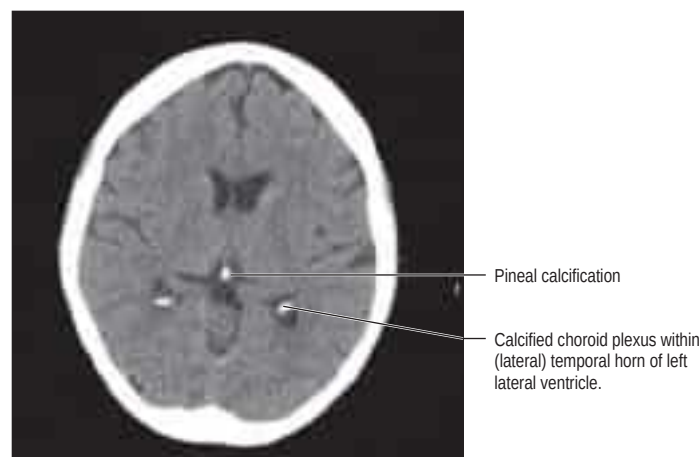


Fig. 23.13 An axial CT image showing calcification of the pineal gland and choroid plexus.

oscillator in the suprachiasmatic nucleus of the hypothalamus governs cyclical pineal behaviour (see above).

From the second decade, calcareous deposits accumulate in pineal extracellular matrix, where they are deposited concentrically as corpora arenacea or 'brain sand' (Fig. 23.13). Pineal gland and choroid plexus calcifications are the most frequent sites of intracranial calcifications incidentally discovered among head computed tomography examinations; calcification can provide a useful indicator of a space-occupying lesion if the gland is significantly displaced from the midline.

Bonus e-book images and table

Fig. 23.5 Magnetic resonance images from four different patients with neurological or psychiatric disorders, showing the position of deep brain-stimulating electrodes.

Table 23.4 Pituitary hormones.

KEY REFERENCES

- Benabid AL, Pollak P, Gervason C et al 1991 Long-term suppression of tremor by chronic stimulation of the ventral intermediate thalamic nucleus. *Lancet* 337:403–6.
The first description of deep brain stimulation of the ventral intermediate (Vim) nucleus of the thalamus for tremor reduction.
- Hodaie M, Wennberg RA, Dostrovsky JO et al 2002 Chronic anterior thalamus stimulation for intractable epilepsy. *Epilepsia* 43:603–8.
A description of the use of deep brain stimulation for seizure reduction in intractable epilepsy.
- Jones EG 1985 *The Thalamus*. New York: Plenum, pp. 403–11.
A description of the nomenclature and connections of thalamic nuclei.
- Lozano AM, Lipsman N 2013 Probing and regulating dysfunctional circuits using deep brain stimulation. *Neuron* 77:406–24.
A review of recent advances in the field of deep brain stimulation.
- Macchi G, Jones EG 1997 Toward an agreement on terminology of nuclear and subnuclear divisions of the motor thalamus. *J Neurosurg* 86: 670–85.
A comparison of the different nomenclatures for motor thalamic nuclei in humans and monkeys and a proposal for a common terminology.
- Macchi MM, Bruce JN 2004 Human pineal physiology and functional significance of melatonin. *Front Neuroendocrinol* 25:177–95.
A review of the neuroanatomy and physiology of the pineal gland, including its rhythmic output functions, particularly the regulation of melatonin synthesis.
- Nieuwenhuys R 1985 *Chemoarchitecture of the Brain*. Berlin: Springer.
A description of the connections and neurochemistry of the hypothalamus.
- Page RB 1982 Pituitary blood flow. *Am J Physiol* 243:E427–42.
A detailed description of the pituitary portal system.
- Sam S, Frohman LA 2008 Normal physiology of hypothalamic pituitary regulation. *Endocrinol Metab Clin North Am* 37:1–22, vii.
A description of the normal physiology of the hypothalamic–pituitary axis.
- Tasker RR, Doorly T, Yamashiro K 1988 Thalamotomy in generalized dystonia. *Adv Neurol* 50:615–31.
A description of the effect of thalamotomy on different aspects of dystonia.

REFERENCES

- Benabid AL, Pollak P, Gervason C et al 1991 Long-term suppression of tremor by chronic stimulation of the ventral intermediate thalamic nucleus. *Lancet* 337:403–6.
The first description of deep brain stimulation of the ventral intermediate (Vim) nucleus of the thalamus for tremor reduction.
- Boccard SG, Pereira EA, Moir L et al 2013 Long-term outcomes of deep brain stimulation for neuropathic pain. *Neurosurgery* 72:221–30.
- Casagrande V, Ichida J 2011 Processing in the lateral geniculate nucleus (LGN). In: Levin LA, Nilsson SFE, Ver Hoeve J et al (eds) *Adler's Physiology of the Eye*, 11th ed. Philadelphia: Elsevier, Saunders; Ch. 29, pp. 574–85.
- Cooper IS 1976 20-year followup study of the neurosurgical treatment of dystonia musculorum deformans. *Adv Neurol* 14:423–52.
- Giacino J, Fins JJ, Machado A et al 2012 Central thalamic deep brain stimulation to promote recovery from chronic posttraumatic minimally conscious state: challenges and opportunities. *Neuromodulation* 15: 339–49.
- Gruber D, Kühn AA, Schoenecker T et al 2010 Pallidal and thalamic deep brain stimulation in myoclonus-dystonia. *Mov Disord* 25: 1733–43.
- Hodaie M, Wennberg RA, Dostrovsky JO et al 2002 Chronic anterior thalamus stimulation for intractable epilepsy. *Epilepsia* 43:603–8.
A description of the use of deep brain stimulation for seizure reduction in intractable epilepsy.
- Hubble JP, Busenbark KL, Wilkinson S et al 1996 Deep brain stimulation for essential tremor. *Neurology* 46:1150–3.
- Jiménez F, Nicolini H, Lozano AM et al 2013 Electrical stimulation of the inferior thalamic peduncle in the treatment of major depression and obsessive compulsive disorders. *World Neurosurg* 80:S30.e17–25.
- Jones EG 1985 *The Thalamus*. New York: Plenum, pp. 403–11.
A description of the nomenclature and connections of thalamic nuclei.
- Lozano AM, Lipsman N 2013 Probing and regulating dysfunctional circuits using deep brain stimulation. *Neuron* 77:406–24.
A review of recent advances in the field of deep brain stimulation.
- Macchi G, Jones EG 1997 Toward an agreement on terminology of nuclear and subnuclear divisions of the motor thalamus. *J Neurosurg* 86: 670–85.
A comparison of the different nomenclatures for motor thalamic nuclei in humans and monkeys and a proposal for a common terminology.
- Macchi MM, Bruce JN 2004 Human pineal physiology and functional significance of melatonin. *Front Neuroendocrinol* 25:177–95.
A review of the neuroanatomy and physiology of the pineal gland, including its rhythmic output functions, particularly the regulation of melatonin synthesis.
- Mohawk JA, Takahashi JS 2011 Cell autonomy and synchrony of suprachiasmatic nucleus circadian oscillators. *Trends Neurosci* 34:349–58.
- Nieuwenhuys R 1985 *Chemoarchitecture of the Brain*. Berlin: Springer.
A description of the connections and neurochemistry of the hypothalamus.
- Page RB 1982 Pituitary blood flow. *Am J Physiol* 243:E427–42.
A detailed description of the pituitary portal system.
- Piedad JCP, Rickards HE, Cavanna AE 2012 What patients with Gilles de la Tourette syndrome should be treated with deep brain stimulation and what is the best target? *Neurosurgery* 71:173–92.
- Sam S, Frohman LA 2008 Normal physiology of hypothalamic pituitary regulation. *Endocrinol Metab Clin North Am* 37:1–22, vii.
A description of the normal physiology of the hypothalamic–pituitary axis.
- Tasker RR, Doorly T, Yamashiro K 1988 Thalamotomy in generalized dystonia. *Adv Neurol* 50:615–31.
A description of the effect of thalamotomy on different aspects of dystonia.
- Xuereb GP, Prichard MM, Daniel PM 1954 The arterial supply and venous drainage of the human hypophysis cerebri. *Q J Exp Physiol Cogn Med Sci* 39:199–217.